Language-Informed Hyperspectral Image Synthesis for Imbalanced-Small Sample Classification via Semi-Supervised Conditional Diffusion Model

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***Abstract*—Data augmentation effectively addresses the imbalanced-small sample data (ISSD) problem in hyperspectral image classification (HSIC). Although most methodologies extend features in the latent space, few leverage text-driven generation to create realistic and diverse samples for balancing limited annotated data. This article proposes Txt2HSI-LDM(VAE), a novel language-informed HSI synthesis method to address the ISSD problem in HSIC. The proposed approach uses a denoising diffusion model, which iteratively removes Gaussian noise to generate hyperspectral samples conditioned on textual descriptions. First, to address the high dimensionality of hyperspectral data, a universal variational autoencoder (VAE) is designed to map the data into a low-dimensional latent space, which provides stable feature representation and significantly reduces the inference complexity of the diffusion model. Second, a semi-supervised diffusion model is designed to fully take advantage of unlabeled data. Random polygon spatial clipping (RPSC) and latent feature uncertainty estimation (LF-UE) without parameters are used to simulate the varying degrees of mixing of training data. Third, the VAE decodes HSI from the latent space generated by the diffusion model with the language conditions as input, contributing to more realistic and diverse samples that match the textual descriptions. In our experiments, we fully evaluate synthetic samples' effectiveness from the aspects of statistical characteristics and data distribution in 2D-principal component analysis (PCA) space. Additionally, the key component, visual–linguistic cross-attention, is visualized on the pixel level to prove that our proposed model can capture the spatial layout and geometry property of the generated HSIs. Experiments carried out on three widely used HSI datasets demonstrate that the proposed Txt2HSI-LDM(VAE) outperforms the classical backbone models, state-of-the-art convolutional neural networks (CNNs), semi-supervised methods, visual–linguistic feature fusion method, as well as diffusion-model-based methods.**

***Index Terms*—Conditional latent diffusion model (LDM), hyperspectral image classification (HSIC), image generation, imbalanced-small sample, semi-supervised learning, visual–linguistic cross-attention.**

I. Introduction

HYPERSPECTRAL images (HSIs) offer rich spatial–spectral information that can help to distinguish various spectrally similar objects to support various environmental applications [1]. Among various hyperspectral processing tasks, HSI classification (HSIC) is a critical one [2], [3], [4], [5], [6], [7], which can support various key application, such as atmospheric environment monitoring [8], [9], [10], geologic mapping [11], [12], and agriculture disaster response [13], [14].

The extraction of spatial–spectral features plays an extremely important role in the field of HSIC. Traditionally, machine learning-based feature extraction methods and morphology-based approaches have been employed [2], [15]. Techniques such as conditional random field (CRF) and Markov random field (MRF) are widely used to integrate spatial context information, enhancing HSIC performance [16]. However, these feature extractors are mostly based on feature engineering, limiting their adaptability to complex data characteristics required for improved HSIC. The data-driven feature learning and neural network (NN)-based approaches, represented by convolutional NNs (CNNs) [17], [18], [19], [20], [21], [22] and transformer models [23], [24], [25], [26], [27], have significantly improved spatial–spectral feature extraction capabilities, thereby greatly boosting HSIC performance.

To leverage unlabeled data for enhanced feature extraction, some self-supervised and unsupervised learning methods have been proposed. These methods employ an encoder–decoder network trained in a self-supervised manner for HSI reconstruction, such as autoencoder-based methods [28], [29], [30], [31], [32] and transfer learning using visual pretraining methods [23], [33], [34]. Another group of hyperspectral processing methods is deep clustering, represented by graph-based approaches [35], [36], [37] through node definition and learning the strength of edge connections to cluster the spatial–spectral features. These models can leverage the label-agnostic multiscale spatial–spectral structure information for enhanced HSIC.

Some unsupervised feature learning methods based on denoising diffusion probabilistic models (DDPMs) have emerged as powerful generative models with superior performance in image generation and restoration tasks [38], [39], [40]. Due to the flexibility of DDPM's architecture and the accuracy of log-likelihood calculation, DDPM can implicitly capture both high-level and low-level visual concepts and model the complicated spatial–spectral relationship via explicit probability criteria [41]. In remote sensing, change

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detection [42], classification [43], [44], [45], [46], [47], and super-resolution [48], [49], [50] tasks have fully explored the characteristic of DDPM. In HSIC, Chen et al. [43] and Zhou et al. [45] utilized a pretrained diffusion model to extract the spatial–spectral diffusion features on denoising U-Net blocks for enhanced HSIC.

However, these unsupervised DDPM-based approaches are limited to feature extractors and cannot be used to generate new data to better address limited training sample problems, especially for small classes that have very limited training samples. The long-tailed and sparse distribution of training data tends to bias the model toward the majority classes [51], [52], [53]. In some specific classification scenes, minority classes are more important from the application perspective, and thereby require a higher classification accuracy than the majority classes, which might be less important [54]. This imbalanced-small sample data (ISSD) problem also leads to a lack of sample diversity, limiting the generalization ability of the algorithm. Hence, it is critical to design new DDPM approaches that can efficiently generate new data and expand training data to deal with the ISSD HSIC problem.

Some data generation approaches have been successfully introduced to address the issue of ISSD in HSIC [55], [56], [57]. Among these, conditional generative adversarial networks (CGANs) are one of the most representative algorithms, which have the capability of generating realistic samples within the class-conditional distribution of real-world data [58], [59]. CGANs employ data augmentation and minority-class oversampling strategies to mitigate ISSD challenges in HSIC [60], [61]. However, a notable limitation of this framework lies in the unstable training mechanism of GANs, which may lead to model collapse and cause the quality degradation of generated samples [62]. Although recent studies have attempted to integrate the minority-class generation and classifier optimization into a unified architecture, the joint optimization of the compound loss function remains a complex task. As Xi et al. [54] reported, due to the high dimensionality of HSI, the class-conditional information might get overshadowed in this approach.

Recently, conditional diffusion models utilizing variational autoencoders (VAEs), known as latent diffusion models (LDMs), have successfully incorporated textual descriptions into the data generation pipeline, significantly enhancing generation performance. Some studies examined the impact of imbalanced class distribution within several long-tailed datasets on diffusion-based generative frameworks [63], [64], demonstrating their unique advantages in addressing minority classes.

Hence, motivated by the success of text-guided LDMs in data simulation, especially for addressing minority classes, we propose a new language-informed HSIC framework named Txt2HSI-LDM(VAE), which integrates the LDM architecture with linguistic guidance to address the ISSD problem in HSIC. Unlike CNNs and other generative models, Txt2HSI-LDM(VAE) embeds language information directly into the framework, ensuring the generated samples align well with both the underlying statistical distribution and the textual descriptions, and enabling the visual fidelity of generated samples while mitigating the cost of extensive manual annotation.

In more detail, the main contributions of this article are summarized as follows.

1) *New HSI Synthesis Framework:* A new HSI synthesis framework based on a language-informed conditional diffusion model, Txt2HSI-LDM(VAE), is proposed to solve the ISSD problem in HSIC. Txt2HSI-LDM(VAE) is a three-stage framework. The first stage uses a VAE to obtain stable representations in low-dimensional latent space, which accelerates diffusion model training and reduces the model's parameters. The second stage utilizes LDM to generate realistic samples under language guidance. The Txt2HSI-LDM(VAE) framework can better address the minority-class issue by generating more balanced data. The third stage combines the synthetic data and real labeled data for enhanced ISSD HSIC.

2) *Reduction of Manual Annotations:* The proposed approach can reduce the burden of making manual annotations by using the fine-grained text description based on the distribution, color, shape, and adjacency relationship of the HSI classes. Moreover, even coarse-grained textual descriptions can improve classification performance. In addition, the language guidance is useful for generating new hyperspectral data in a language-guided manner for different purposes.

3) *Improved Model Generalization Capability:* The proposed approach can improve the model's generalization capability, because, at the second stage, we design a semi-supervised LDM approach to fully leverage the unlabeled data. Additionally, to better capture the underlying statistical distribution, random polygon spatial clipping (RPSC) and latent feature uncertainty estimation (LF-UE) are used to simulate the spatial–spectral heterogeneity and spectral mixing phenomena in real HSI data.

4) *Enhanced Classification Model:* An HSI classification approach is used that can make full use of both real and simulated data to address the ISSD problem in HSIC.

The remainder of this article is organized as follows. Section II briefly introduces the related works. Section III describes our proposed method in detail. Section IV shows the experimental results. Section V draws the conclusion and discussion.

II. Related Work

A. ISSD HSIC

Several prior works, including data augmentation, resampling (oversampling and undersampling) [65], [66], deep transfer learning [23], [67], pseudo-label learning [68], and deep few-shot learning [69], have been proposed to address the ISSD HSIC problem. For instance, random oversampling (ROS) and random undersampling (RUS) simply replicate the training samples, which leads to data redundancy [70]. Synthetic minority oversampling technique (SMOTE) may contribute to better classification performance on conventional classifiers, but this method uses a simple linear combination of one random sample and its nearest neighbors to generate synthetic minority samples, which fails to capture the complex manifold structure and is hard to satisfy the nonlinear spatial–spectral relationship in the HSI data [71].

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Latent variable models, especially represented by the deep generative models (DGMs), introduce uncertainty into the low-dimensional discriminative latent space during the reconstruction process, possessing the ability to stimulate the complex heterogeneity of HSI data and generate samples for the minority class. Particularly, GANs replace the complexity of resampling by searching for a Nash equilibrium of the generator and discriminator through adversarial training. Specifically, class-informed mixture GANs are developed to ensure the generated samples align with the actual data distribution. For example, Dam et al. [60] proposed a parallel mixture generator spectral 1D-GAN to generate class-dependent samples by fully exploiting the spectral feature. Auxiliary classifiers significantly improved the classification performance of imbalanced datasets. Zhu et al. [19] proposed a 3D-GAN that took both spatial and principal component analysis (PCA)-reduced spectral features into account to handle 3-D patch samples. The discriminator not only distinguished the real data from the synthetic fake data but also leveraged the softmax classifier to produce the classification maps.

The common deep transfer learning methods mainly use some pretrained models in the natural images field, such as VGG16 and ResNet50 with frozen parameters, to fine-tune the downstream tasks in the remote sensing community. Thanks to these pretrained models, they can obtain promising results with a small sample size and lower computational burden. Jiao et al. [33] proposed a deep multiscale feature extraction algorithm, which fully took advantage of pretrained VGG-16, to exploit the spatial structure information of HSIs by performing a full convolution operation, and then realized adaptive spatial–spectral information fusion by employing a weighted strategy. Wang et al. [34] first built three-spectral band hyperspectral datasets through a random band selection (RBS) procedure, letting VGG-16 learn the feature of homogeneous areas with distinguished semantic and geometric properties. However, extracting features of HSI with models that are pretrained on natural RGB images requires generalizing to hundreds of bands from triple-band datasets, which is a challenging task.

B. Image Synthesis for HSIC

To estimate the data distribution for more realistic image synthesis and expand the small training data, scholars have developed several methods, which have achieved promising results. The commonly used algorithms include the mixture model (Gaussian mixture model, GMM), generative model (GAN and VAE), and DDPM.

Deep learning-based methods, with GAN and VAE being typical representatives, generate new samples of a given distribution. Specifically, this is achieved by training a generator to map random noise from the latent space to the data distribution. Instead of inferring the global distribution, scholars employed conditional generative models, which are designed to condition the output of the generator based on the hyperspectral classes. For instance, Audebert et al. [72] used a CGAN to generate an arbitrarily large number of hyperspectral samples that matched the distribution of any dataset. Considering spectral mixing, this method can synthesize any combination of classes by interpolating between vectors in the latent space. Due to the high-dimensional HSI data with complex noise, VAE is more suitable for processing HSI by introducing the Gaussian noise into the encoded results in the latent space. Xi et al. [54] proposed a deep generative spectral-spatial classifier (DGSSC) method, which can achieve minority-class data augmentation in the latent variable space to address the problem of imbalanced data. However, the condition was only based on one-hot encoding of hyperspectral classes. Considering the complex situations in the real-world application and the semantic context, it is likely that the one-hot-like encoding in probabilistic form, such as pixel abundance, instead of a binary vector, will work better.

Very recently, DDPMs have demonstrated superior performance in image inpainting and synthesis compared to GANs and VAEs. For instance, Zhang et al. [73] proposed RGB images to hyperspectral image conditional cascade diffusion (R2HCCD), an algorithm for HSI generation that leveraged the conditional DDPM to produce high-fidelity HSI from an RGB image through cascaded generation. Liu et al. [74] proposed a diverse hyperspectral remote sensing image generation method based on LDM (HyperLDM), employing a conditional vector quantized GAN (VQGAN) to compact high-dimensional information for accelerating the diffusion process. Additionally, HyperLDM incorporated abundance maps derived from VQGAN to condition the HSI generation via LDM. By doing so, the model effectively accounted for the spectral mixing of multiple materials within HSI pixels. However, a critical limitation of HyperLDM is its reliance on an accurately calibrated spectral library during the transformation of abundance maps into HSI using the linear mixture model (LMM).

However, the above methods, whether CGANs, conditional VAEs, or latent conditional DDPMs, still require substantial quantities of paired remote sensing image observations (e.g., RGB image and corresponding HSI). This poses a significant challenge in practical applications, due to the limited availability of such paired datasets arising from satellite revisit period variations and heterogeneous sensor specifications across different observation platforms.

C. Image–Text Pairing Information in HSIC

HSI analysis demands intensive manual labeling, while language representations are inherently more accessible and straightforward for humans to express and acquire. Recent advancements in artificial intelligence (AI) methods, such as contrastive language image pretraining (CLIP) [75], have demonstrated the capability to model the bidirectional relationships between these two modalities (image and text). Moreover, these methods present exceptional transferability across diverse downstream visual and language tasks, making them promising candidates for addressing the data scarcity challenges in HSI-related applications [76], [77].

Zhang et al. [78] proposed a language-aware domain generalization network (LDGNet) for HSIC. This method learned visual–linguistic alignment features from source datasets and transferred cross-domain invariant knowledge to target domains. Similarly, Dang et al. [79] developed LIVEnet for

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cross-scene HSI classification, which enhanced the information exchange between linguistic and visual features while exploiting interclass correlation prevalent in imbalanced sample scenarios. Moreover, Cao et al. [80] designed a three-aspect text description (color, label, and shape) to extract linguistic prior information, enabling shared spatial and spectral features in a unified semantic space. Given these successes, however, the text-guided approaches have not been fully explored for addressing the ISSD problem in HSIC.

D. Latent Conditional Diffusion Model

Diffusion models [40] are probabilistic models designed to learn data distributions $p(x)$ by gradual denoising a normally distributed variable, which corresponds to learning the reverse process of a fixed Markov chain of total length T.

However, for high-dimensional distributions, diffusion model training [see (1)] and sampling [see (2)] require massive computational resources

$$\mathcal{L}_{\mathrm{DM}} = \mathbb{E}_{x,\epsilon\sim\mathcal{N}(0,1),t}\left[\|\epsilon - \epsilon_\theta(x_t,t)\|_2^2\right] \tag{1}$$

$$q(x_{t-1}|x_t,t) = \mathcal{N}\left(x_{t-1}; \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t,t)\right), \beta_t\mathbf{I}\right) \tag{2}$$

where $\alpha_t =: 1-\beta_t$, $\bar{\alpha}_t =: \prod_{s=1}^{t}\alpha_s$, $\epsilon \sim \mathcal{N}(0,\mathbf{I})$, and ϵ_θ represents the diffusion model with parameter θ.

To enhance the efficiency of the training and sampling process, Rombach et al. [81], the authors of stable diffusion, proposed LDMs, which performed the diffusion in the latent space of a pretrained VAE. Specifically, given a frozen encoder $\mathcal{E}: \mathbb{R}^n \to \mathbb{R}^k$ and decoder $\mathcal{D}: \mathbb{R}^k \to \mathbb{R}^n$, train a diffusion model of representations $z = \mathcal{E}(x_0)$, where x_0 represents the original data. The noisy samples can be created by

$$q(z_t|\mathcal{E}(x_0),t) = \mathcal{N}\left(z_t; \sqrt{\bar{\alpha}_t}\mathcal{E}(x_0), (1-\bar{\alpha}_t)\mathbf{I}\right). \tag{3}$$

The corresponding objective can be written as

$$\mathcal{L}_{\mathrm{LDM}} = \mathbb{E}_{z\sim\mathcal{E}(x),\epsilon\sim\mathcal{N}(0,1),t}\left[\|\epsilon - \epsilon_\theta(z_t,t)\|_2^2\right]. \tag{4}$$

At inference time, new images can then be generated by sampling a clean representation z from the diffusion model and subsequently decoding it to an image with the learned decoder $x = \mathcal{D}(z)$.

In principle, diffusion models have the capability to formulate conditional distributions in the form of $p(z|c)$. This can be implemented with a conditional denoising autoencoder $\epsilon_\theta(z_t,t,c)$, which paves the way to control the synthesis process through inputs c, such as text [82], semantic maps [83], and other image translation tasks [84].

III. Proposed Txt2HSI-LDM(VAE)

A. Overview of the Proposed Txt2HSI-LDM(VAE)

In this section, we introduce the Txt2HSI-LDM(VAE) framework, which integrates the strengths of VAE (for dimension reduction) and language-informed conditional diffusion models, Txt2HSI-LDM for short, to address the ISSD problem in HSIC by generating realistic HSI synthesis. The proposed method is specifically designed to enhance classification performance, particularly when the labeled dataset is imbalanced and small. As shown in Fig. 1, the framework consists of three main phases: 1) VAE pretraining; 2) Txt2HSI-LDM training; and 3) Txt2HSI-LDM inference and classification. It is worth noting that, in the second phase, to expand and enrich the training data, Txt2HSI-LDM is trained in a semi-supervised fashion using abundant unlabeled data. In order to further enhance the diversity of the sample, two methods without parameters, RPSC and LF-UE, are used to stimulate the diversity of HSIs. This architecture enables us to harness the information compression capabilities of VAE and the generative potential of conditional diffusion models to address ISSD challenges.

B. Latent Space Representation via VAE

As depicted in Section II-D, the first stage of the proposed model Txt2HSI-LDM(VAE) is implemented with VAE to obtain compressed and stable feature representations in latent space. The main components include four parts: an encoder network, a reparameterization process, a decoder network, and a discriminator network (see Fig. 2 for detailed architectures of the encoder and decoder networks). Given a hyperspectral instance $\mathcal{H} \in \mathbb{R}^{C\times H\times W}$, where H and W are the height and width of the hyperspectral patch, respectively, and C is the number of spectral bands, the encoder of VAE projects the $\mathcal{H}$ into $z \in \mathbb{R}^{4\times H\times W}$, which is modeled by $\boldsymbol{q}_{\mathcal{E}_\theta}(\boldsymbol{Z}|\boldsymbol{\mathcal{H}})$. The reconstructed hyperspectral instance $\mathcal{H}'$ can be obtained by feeding the reparameterized $\boldsymbol{Z}$ to the decoder part, modeled by $\boldsymbol{p}_{\mathcal{D}_\theta}(\boldsymbol{\mathcal{H}}|\boldsymbol{Z})$. Reparameterization is a process that samples a Gaussian distribution with mean $\mu(Z)$ and $\Sigma(Z)$ variance of latent feature Z.

We optimize the VAE under the framework of GANs to make the reconstruction of HSI more realistic. The discriminator is the same as in our previous work [85]. The VAE is optimized by minimizing the following objective function:

$$\begin{aligned}
&\min_{\theta_\mathcal{E},\theta_\mathcal{D}}\max_{\theta_\mathcal{G}}\left(\mathcal{L}_{\mathrm{vae}} + \mathcal{L}_{\mathrm{adv}}\right)\\
&\mathcal{L}_{\mathrm{vae}} = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\|\mathcal{H}'_{ij} - \mathcal{H}_{ij}\|_2^2\\
&\qquad + \lambda_{\mathrm{KL}}\frac{1}{N}\sum_{i=1}^{HW} D_{\mathrm{KL}}\left(q_{\mathcal{E}_z}(z_i|\mathcal{H}_i)\,\|\,p(z)\right)\\
&\mathcal{L}_{\mathrm{adv}} = \mathbb{E}_{\mathcal{H}\sim p(\mathcal{H})}\log\mathbf{Dis}(\mathcal{H})\\
&\qquad + \mathbb{E}_{\mathcal{H}\sim p(\mathcal{H})}\log\left[1 - \mathbf{Dis}\left(\mathcal{H}'\right)\right]
\end{aligned} \tag{5}$$

where $\theta_\mathcal{E}$, $\theta_\mathcal{D}$, and $\theta_\mathcal{G}$ denote the parameters in encoder $\mathcal{E}$, decoder $\mathcal{D}$, and discriminator **Dis**, respectively. $q_{\mathcal{E}_z}(z_i|\mathcal{H}_i)$ is the variational distribution in the VAE.

C. Preliminary of Diffusion Model

The diffusion model is a latent variable model that contains a forward diffusion process and a reverse diffusion process. Both forward and reverse processes can be defined as a parameterized Markov chain, and the reverse process can estimate the added noise in the forward diffusion process for generation [40].

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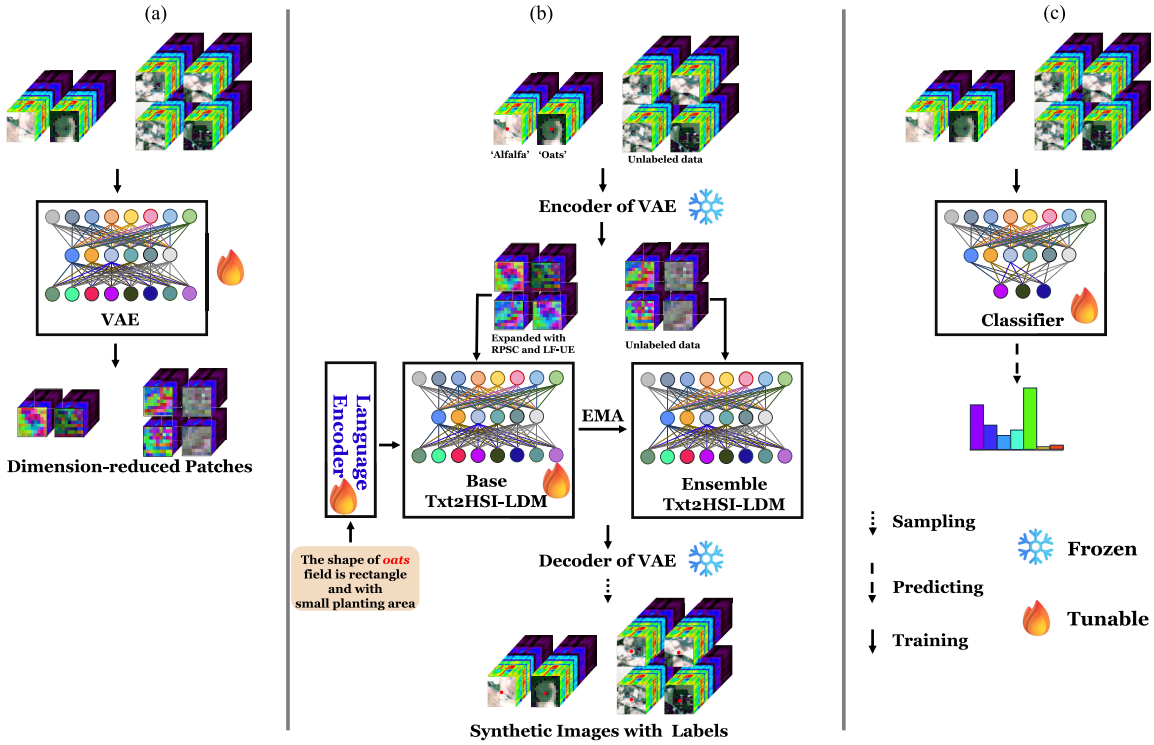


Fig. 1. Overview of Txt2HSI-LDM(VAE). First, a VAE is trained on all the cropped patch HSI data, and the dimension-reduced data are used for the second stage. Second, a language-informed conditional diffusion model, Txt2HSI-LDM, is trained on limited labeled data and unlabeled data in a semi-supervised way to generate synthetic images given random language descriptions. Finally, the classifier [22] is trained or fine-tuned on ISSD data expanded by generated images with labels. EMA means exponential moving average to transfer parameters from the base model to the ensemble model. The language encoder uses the pretrained parameter of the CLIP model named "ViT-B-32.pt" [75], and we fine-tune it together with the diffusion model. (a) All cropped patches. (b) Partially labeled real HIS. (c) ISSD expanded by generated ones.

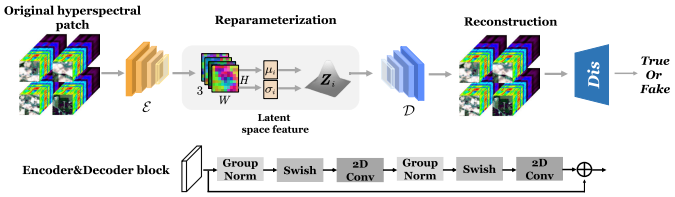


Fig. 2. Illustration of the VAE, which consists of an encoder network, a decoder network, a reparameterization part, and a discriminator network. The detailed architectures of the encoder and decoder blocks are also depicted.

1) Forward Diffusion Process: We set the latent space feature learned by the encoder $\mathcal{E}$ of VAE before adding the Gaussian noise as z_0. The forward process is inspired by nonequilibrium thermodynamics, which can be viewed as a Markov chain that the noise is gradually added to z_0 to produce latent variables z_1 through z_T in T timesteps with fixed linear variance schedule β_t. At timestep t, the noisy feature z_t can be represented as follows:

$$q(z_1, z_2, \ldots, z_T|z_0) = \prod_{t=1}^{T} q(z_t|z_{t-1}) \tag{6}$$

$$q(z_t|z_{t-1}) = \mathcal{N}\left(z_t;\ \sqrt{\beta_t - 1}z_{t-1}, \beta_t \mathbf{I}\right). \tag{7}$$

Through the reparameterization trick, z_t can be sampled directly at an arbitrary time step t in closed form, and the marginal distribution of intermediate latent variables z_t given z_0 can be derived as

$$q(z_t|z_0) = \mathcal{N}\left(z_t;\ \sqrt{\bar{\alpha}_t}z_0, (1-\bar{\alpha}_t)\mathbf{I}\right) \tag{8}$$

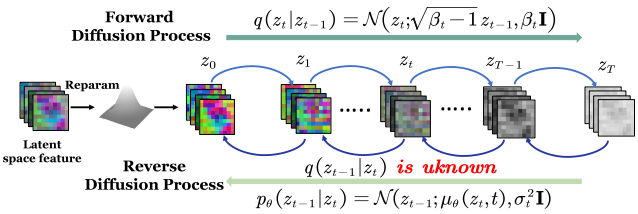


Fig. 3. Latent diffusion forward and backward processes. $q(z_t|z_{t-1})$ and $p_\theta(z_{t-1}|z_t)$ represent the noise-adding forward process and denoising backward process, respectively. The essential question is to estimate the conditional probability $q(z_{t-1}|z_t)$. z_T is nearly pure Gaussian noise.

$$z_t = \sqrt{\bar{\alpha}_t}z_0 + (1-\bar{\alpha}_t)\,\epsilon \tag{9}$$

where $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^{t} a_s$, and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. In (7), the magnitude of the added Gaussian noise decreases as the value of α_t increases, which also means that the informative of z_0 decreases as the time step t increases in (9). When $t = T$, z_T will become pure Gaussian noise.

2) Reverse Diffusion Process: In the reverse process (Fig. 3), the estimation of $q(z_{t-1}|z_t)$ is intractable and depends on the whole data distribution, and a denoising NN ϵ_θ is used to estimate it, where θ represents the network parameters, and then, we generate the real sample from Gaussian sample $z_T \sim \mathcal{N}(0, \mathbf{I})$ with conditional text constraint c, motivated by classifier-free diffusion guidance [86]. According to the property of Markov chain, (6), and (7), the reverse distribution

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p_θ can be formulated as follows:

$$p_\theta(z_{0:T}|c) = p(z_T)\prod_{t=1}^{T} p_\theta(z_{t-1}|z_t, c) \tag{10}$$

$$p_\theta(z_{t-1}|z_t, c) = \mathcal{N}(z_{t-1}; \mu_\theta(z_t, t, c), \Sigma_\theta(z_t, t, c)). \tag{11}$$

We set $\Sigma_\theta(z_t, t, c) = \sigma_t^2\mathbf{I} = (1-\bar{\alpha}_{t-1}/1-\bar{\alpha}_t)\beta_t\mathbf{I}$ to untrained time-dependent constants. For $\mu_\theta(z_t, t, c)$, the conditional distribution of $p_\theta(z_{t-1}|z_t, c)$ can be reparameterized as follows:

$$\mu_\theta(z_t, t, c) = \frac{1}{\sqrt{\alpha_t}}\left(z_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(z_t, t, c)\right). \tag{12}$$

Finally, the objective function of the diffusion model can be constructed by the following equation, and the detailed solutions and derivation process can be referred to [40]:

$$\mathcal{L}_{\mathrm{DM}}(\theta) = \mathbb{E}_{z_0,c,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t, c\right)\right\|_2^2\right]. \tag{13}$$

3) Sampling Process: In the inference process, which is also known as the sampling process, a new latent feature z_0 can be generated from either Gaussian noise or a noisy image z_t by iteratively sampling z_{t-1} until $t = 1$ according to the following equation. We adopt denoising diffusion implicit models (DDIMs) [87] to accelerate the sampling process

$$z_{t-1} = \sqrt{\alpha_{t-1}}\left(\frac{z_t - \sqrt{1-\alpha_t}\bar{\epsilon}_\theta}{\sqrt{\alpha_t}}\right) + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\bar{\epsilon}_\theta + \sigma_t\epsilon_t \tag{14}$$

where $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$. Following classifier-free guidance diffusion [86], we set $\bar{\epsilon}_\theta = (1+\omega)\epsilon_\theta(z_t, t, c) - \omega\epsilon_\theta(z_t, t, c = \oslash)$. $c = \oslash$ is done by randomly dropping out c during training and replacing it with a learned "null" embedding $\oslash$. Generally, training with classifier-free guidance requires two models: an unconditional generation model and a conditional generation model. However, these two models can be unified into a single model by probabilistically omitting the language condition during training. During inference, the final result can be achieved by linear extrapolation between the conditional and unconditional generations. This allows for adjustment of the generation effect to balance the fidelity and diversity of the generated samples by changing the guidance coefficient ω.

D. Conditional Denoising Diffusion in the Latent Space

The hyperspectral data usually have low spatial resolution, follow a long-tail distribution, and have high-dimensional properties. To speed up the generation process of the diffusion model and reduce the number of parameters, we perform diffusion in the latent space to cope with the high dimensionality and uneven range of the hyperspectral data. Simultaneously, if we use the whole-image level hyperspectral data as the input of the VAE encoder, predictably, the step-by-step spatial downsampling operations will lose the detail of images, which leads to negative effects on the generated samples, as well as the classification results. Hence, we crop hyperspectral data into patches and encode these into a latent space using a universal VAE, which only reduces the spectral dimension and keeps the spatial size. Then, we train a conditional LDM in this space to generate latent features with conditional fine-grained

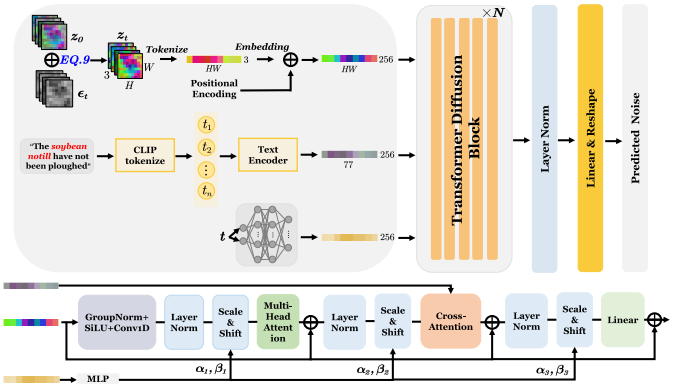


Fig. 4. Illustration of the (top) transformer diffusion network and (bottom) details of blocks.

text descriptions as input. Finally, we decode the latent features into HSI using the VAE decoder. An overview of our proposed model, Txt2HSI-LDM(VAE), is shown in Fig. 1.

1) Noise Schedule: A small enough a_T guarantees a convergent to a normal distribution as the sampling process start value. The total time step T is set to 500. Inspired by [88], we take a linear schedule with minimum 0.0001 and maximum 0.02, where $\bar{\alpha}_t = 4.04 \times 10^{-5}$. As we can see from (9), when $t = T$, z_T hardly contains the information of z_0 and can be seen as normal distribution. Hence, we can start the denoising process with a randomly sampled noise from the normal distribution.

2) Overview of the Proposed Language-Informed Diffusion Model: The detail of architecture can be seen in Fig. 4. Txt2HSI-LDM(VAE) has three kinds of input, i.e., noised latent feature z_t, corresponding time step t, and text description l. For unifying the dimension of the input tensor, and adapting to the input of the transformer diffusion block (TDB), we set all the dimensions of embedded features d_{emb} (including latent feature, embedded textual description, and timesteps) before feeding into the TDB as 512. Similar to the transformer model, for both text and latent feature tokens, we use an embedding layer with positional encoding to transform them into the embedding space with a dimension of d_{emb}, which can help to inject the relative or absolute position information among different tokens in the sequence. For time-step embedding, we use multilinear layers to project discrete-time information into continuous space with a dimension of d_{emb}. Let $c_l \in \mathbb{R}^{n\times d_{\mathrm{emb}}}$, $z_t^{\mathrm{emb}} \in \mathbb{R}^{HW\times d_{\mathrm{emb}}}$, and $t_{\mathrm{emb}} \in \mathbb{R}^{d_{\mathrm{emb}}}$ denote the obtained text embedding, latent feature embedding, and time embedding, respectively. H and W represent the spatial size of hyperspectral patch data, and n represents the total length of the text tokens, here, $n = 77$. TDB is mainly built on multihead attention (MHA), cross-attention, and skip-connection. The t_{emb} is used to rescale the feature with scale and shift before the skip connection. We project t_{emb} to $\alpha_i, \beta_i, i = 1, 2, and\ 3$, as shown in the following equation. The input feature will be mapped multiplied by α_i and added by β_i

$$\alpha_i, \beta_i = \mathrm{Chunk}(\mathrm{MLP}(t_{\mathrm{emb}})) \tag{15}$$

where Chunk is splitting the feature into two parts with equal size along the channel dimension. MLP is implemented as

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an sigmoid linear unit (SiLU) activation followed by a linear layer.

3) Text Encoder: First, we need to justify the text description rules used in this article. We design two different kinds of descriptions, i.e., coarse-grained and fine-grained. The coarse-grained description is constructed by the template of "A hyperspectral image of + *class name*," such as "A hyperspectral image of trees." While the fine-grained text description is manually designed based on the distribution, color, shape, and adjacency relationship [78], for example, "*Alfalfa* occupies a small area and appears light green," "The *shadows* are next to the buildings and appear black color." This solution is widely used in multisource (image and language models) as well as large language model (LLM) inspired remote sensing foundation model, such as RSGPT [89] and LIVEnet [79]. In RSGPT, the language for large-scale remote sensing scene images is manually designed by experts, containing the scene theme, shape, color, quantity, and position (relative and absolute). The first article that discussed the language generated by machines should also consider these five aspects [90]. These two studies focus on the large-scale remote sensing scene captioning, which is not suitable for patch-level HSIC. Although using LLMs can generate detailed descriptions, remote sensing texts often contain technical terms (such as "NDVI anomaly area"), which cannot be decoded by LLMs, such as DeepSeek and ChatGPT, which are pretrained on natural languages. We believe that the powerful performance of RSGPT [89] is based on the refined manual annotation, costing a total of 35 person days, involving five remote sensing experts. Another example [91] can support that we prefer to rely on manually crafted text descriptions. Das et al. [91] built HSI land-cover captioning datasets by automatic generation and manual refinement, but the generated texts were more likely to be general characteristics, for example, "*Alfalfa: Dense leafy canopy with soft tones across wide patches*." Hence, manual annotations make sure that we can also take into account the general topological relations between ground cover types while keeping the whole image's spatial features.

The text encoder is a 512-wide language-model transformer [75] with three layers and eight attention heads. First, the text tokens are obtained by a lower cased byte pair encoding (BPE). The maximum sequence of text length is limited to 77, and we use 0 as the placeholder to fill the empty tokens because the number of words in the input text description is usually smaller than 77. This linguistic feature with positional encoding is then normalized by layer and linearly projected into the latent space.

4) Visual–Linguistic Cross-Attention: Given the property that the attention mechanism can realize the multimodality data fusion [92], [93], hence, we design the cross-attention to fuse to information of text embedding feature and latent feature embedding to ensure its supervision of language on the generation process. The detail of cross-attention is shown in Fig. 5. The value of cross-attention is computed by the following equation:

$$\mathrm{CA} = \mathrm{Attention}\left(\mathcal{Q}_{\mathrm{img}}, \mathcal{K}_{\mathrm{txt}}, \mathcal{V}_{\mathrm{txt}}\right) = \sigma\left(\frac{\mathcal{Q}_{\mathrm{img}}\mathcal{K}_{\mathrm{txt}}^{\mathrm{T}}}{\sqrt{d_{\mathrm{emb}}}}\right)\mathcal{V}_{\mathrm{txt}} \quad (16)$$

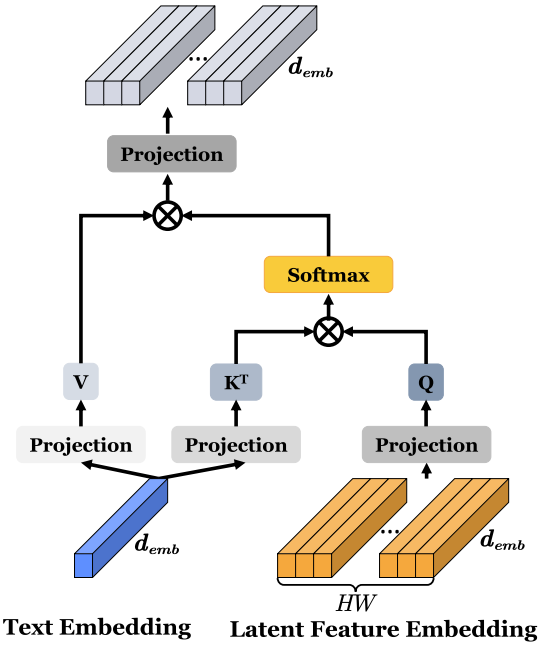


Fig. 5. Illustration of the cross-attention.

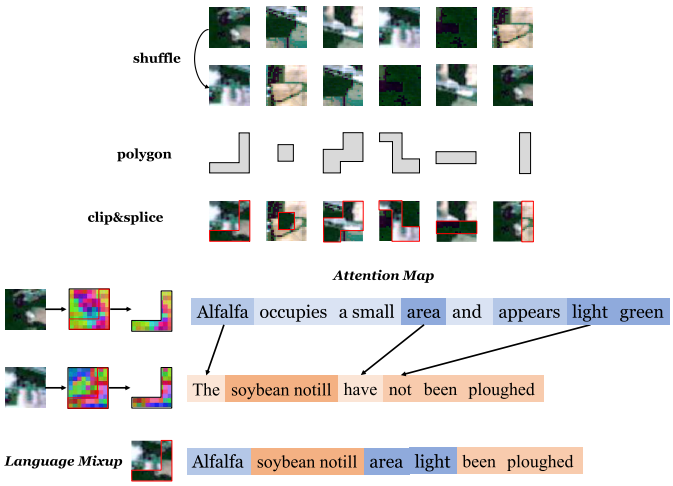


Fig. 6. Illustration of the RPSC and language mixup. The deeper color means the higher attention value.

where $\mathcal{Q}_{\mathrm{img}} = \mathcal{F}_{\mathrm{img}}\mathcal{W}_{\mathrm{img}}^{\mathcal{Q}}$, $\mathcal{K}_{\mathrm{txt}} = \mathcal{F}_{\mathrm{txt}}\mathcal{W}_{\mathrm{txt}}^{\mathcal{K}}$, and $\mathcal{V}_{\mathrm{txt}} = \mathcal{F}_{\mathrm{txt}}\mathcal{W}_{\mathrm{txt}}^{\mathcal{V}}$ are the queries, keys, and values, respectively, and $\mathcal{W}_{\mathrm{img}}^{\mathcal{Q}}$, $\mathcal{W}_{\mathrm{txt}}^{\mathcal{K}}$, and $\mathcal{W}_{\mathrm{txt}}^{\mathcal{V}}$ represent the projection weights of $\mathcal{Q}_{\mathrm{img}}$, $\mathcal{K}_{\mathrm{txt}}$, and $\mathcal{V}_{\mathrm{txt}}$, respectively. $\mathcal{F}_{\mathrm{img}}$ and $\mathcal{F}_{\mathrm{txt}}$ represent the feature of image and text, respectively. σ indicates the *softmax* function.

5) Semi-Supervised Denoising Network and Loss Function: For better dealing with the ISSD problem in HSIC, and considering the sparse labeled data in the real world, one of the effective and promising methods is to fully learn knowledge from the unlabeled data and facilitate information exchange between the knowledge of unlabeled and labeled data in a way of knowledge distillation. Semi-supervised learning framework can realize this kind of purpose. Some related works and contents can be found in [94], [95], [96], and [97].

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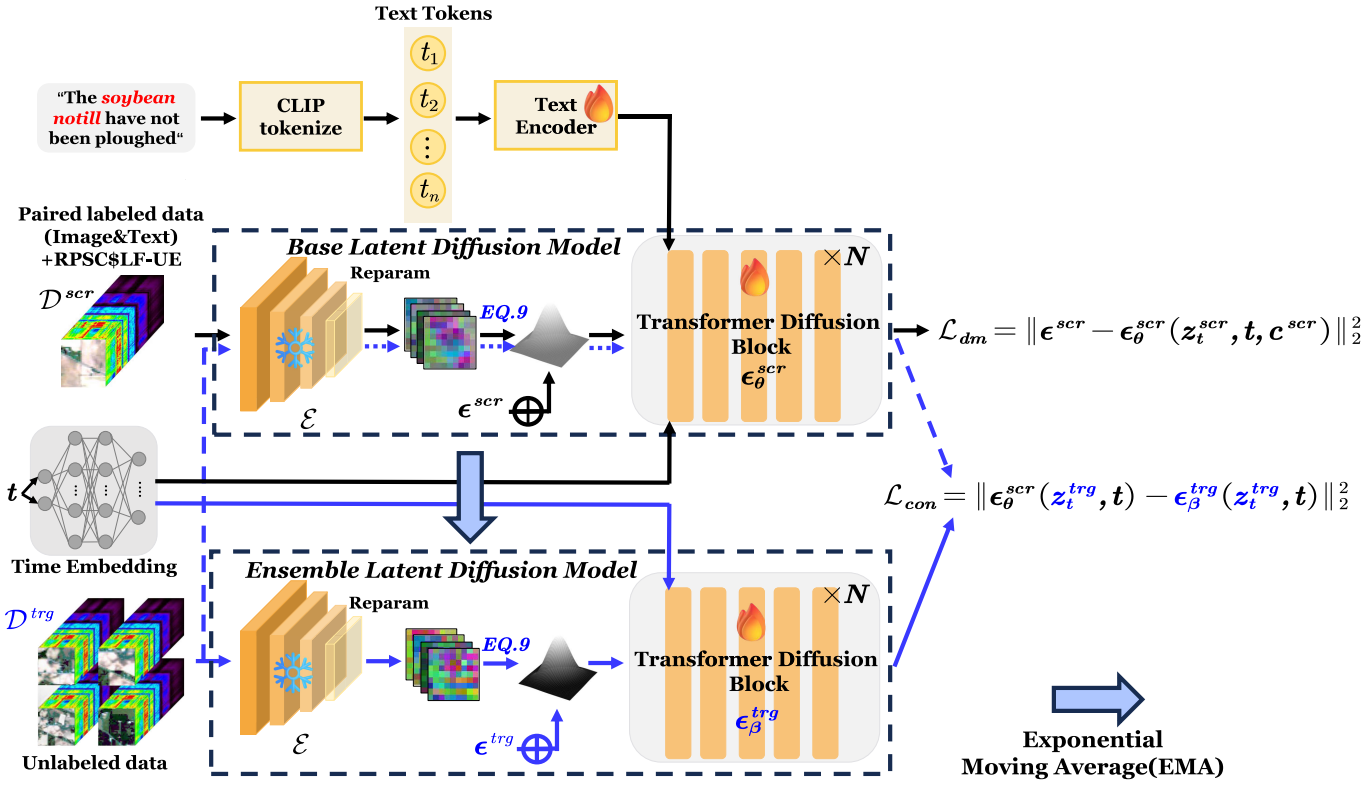


Fig. 7. Detail of the semi-supervised learning part. In the top part, the text descriptions are first converted into tokens, and then, the vectors are encoded by the on-hand CLIP pretrained weight "ViT-B-32.pt." On the bottom part, a bunch of unlabeled data are, respectively, fed into base LDM $\epsilon_\theta^{\text{scr}}$ and ensemble LDM $\epsilon_\beta^{\text{trg}}$.

Additionally, for better simulating the spectral mixing and heterogeneity and expanding the diversity of training data, we introduce two kinds of nonparameter methods, i.e., RPSC and LF-UE. The process of RPSC is depicted in Fig. 6, and the language description of the clipped sample is generated by mixing the attention value of data before clipping [98]. The generalized sample of latent space feature z obtained by LF-UE is formulated by the following equation:

$$\begin{aligned}\phi(z) &= \mu(z) + \epsilon_\mu \Sigma_\mu(z), \quad \epsilon_\mu \sim \mathcal{N}(0, \mathbf{I}) \\ \varphi(z) &= \sigma(z) + \epsilon_\sigma \Sigma_\sigma(z), \quad \epsilon_\sigma \sim \mathcal{N}(0, \mathbf{I}) \\ \tilde{z} &= \phi(z) + \varphi(z) \times \frac{z - \mu(z)}{\sigma(z)}\end{aligned} \tag{17}$$

where $\Sigma_\mu(z)$ and $\Sigma_\mu(z)$ are the batch variance of the spatial mean $\mu(z)$ and standard deviation (std) $\sigma(z)$, respectively. To model uncertainty, Gaussian noise is injected into the above statistics, and the final perturbed latent feature $\tilde{z}$ is finally obtained. As shown in Fig. 7, $\mathcal{D}^{\text{scr}}$ and $\mathcal{D}^{\text{trg}}$ represent the limited labeled patch data with corresponding fine-grained text descriptions and a bunch of unlabeled patch data, respectively. On the top part of Fig. 7, the CLIP tokenization returns the tokenized representation of given text input l, for example, "the soybean notill fields have not been ploughed" will be converted into a 2-D tensor [**49 406**, 518, 34 606, 2534, 660, 720, 783, 1025, 33 811, 538, **49 407**, 0, 0,..., 0] $\in \mathbb{R}^{77}$. Next, the tensor will be encoded by an on-hand pretrained text encoder $\mathcal{T}_\phi$ as the condition of base latent diffusion $\epsilon_\theta^{\text{scr}}$. On the bottom part of Fig. 7, abundant unlabeled patch data randomly selected are fed into the unconditional ensemble LDM $\epsilon_\beta^{\text{trg}}$ with parameter β before being encoded by VAE. Both $\epsilon_\theta^{\text{scr}}$ and $\epsilon_\beta^{\text{trg}}$ share the same architecture. During the training phase, the diffusion loss is calculated on the labeled sample with the following equation:

$$\mathcal{L}_{\text{dm}}\left(\theta, \phi, z_t^{\text{scr}}, c^{\text{scr}}\right) = \left\| \epsilon^{\text{scr}} - \epsilon_\theta^{\text{scr}}\left(z_t^{\text{scr}}, t, c^{\text{scr}}\right) \right\|_2^2 \tag{18}$$

where $c^{\text{scr}} = \mathcal{T}_\phi(\text{tokenize}(l))$, ϕ is the weight of text encoder $\mathcal{T}$, and we optimize ϕ together with $\epsilon_\theta^{\text{scr}}$, while the unlabeled samples are fed into both the base LDM and the ensemble LDM to calculate the diffusion consistency loss with mean-squared error as

$$\mathcal{L}_{\text{con}}\left(\theta, z_t^{\text{trg}}\right) = \left\| \epsilon_\theta^{\text{scr}}\left(z_t^{\text{trg}}, t\right) - \epsilon_\beta^{\text{trg}}\left(z_t^{\text{trg}}, t\right) \right\|_2^2. \tag{19}$$

At each iteration, the $\epsilon_\theta^{\text{scr}}$ is updated with the gradients from both two losses. The $\epsilon_\beta^{\text{trg}}$ does not participate in this process. Instead, its parameters are manually updated using the exponential moving average (EMA) of the historical parameters in the $\epsilon_\theta^{\text{scr}}$ to realize knowledge distillation. At the τth iteration, β^τ can be updated by

$$\beta^\tau = \alpha \beta^{\tau-1} + (1 - \alpha)\theta^\tau \tag{20}$$

where α is a smoothing coefficient, and we set to 0.99. Our goal is to let the $\epsilon_\theta^{\text{scr}}$ learn knowledge from the $\epsilon_\beta^{\text{trg}}$ on unlabeled data. Therefore, the parameters in $\epsilon_\beta^{\text{trg}}$ does not participant the optimization. The full objective function for training the base latent diffusion mode $\epsilon_\theta^{\text{scr}}$ can be formulated as

$$\min_{\theta, \phi} \left(\mathcal{L}_{\text{dm}}\left(\theta, z_t^{\text{scr}}, c^{\text{scr}}\right) + \mathcal{L}_{\text{con}}\left(\theta, z_t^{\text{trg}}\right) \right). \tag{21}$$

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Algorithm 1 Training of the Semi-Supervised LDM Txt2HSI-LDM(VAE)

Input: labeled data $\mathcal{D}^{scr}$ with corresponding text description l, unlabeled data $\mathcal{D}^{trg}$, learning rate η, total epoch E, initialized parameter θ, β, ϕ for $\epsilon^{scr}, \epsilon^{trg}, \mathcal{T}$ respectively

Output: The trained model ϵ^{scr} with parameter θ

while $e < E$ **do**
  Sample $\mathcal{D}^{scr} \sim q(\mathcal{D}^{scr})$, $\mathcal{D}^{trg} \sim q(\mathcal{D}^{trg})$,$\epsilon^{scr} \sim \mathcal{N}(0, \mathbf{I})$, $t \sim U(0, \ldots, T)$, $p \sim U(0, 1)$
  Feed $\mathcal{D}^{scr}$ and $\mathcal{D}^{trg}$ into VAE encoder and get z^{scr}, z^{trg} respectively
  Encode the language using $\mathcal{T}$ with parameter ϕ and get c^{scr}
  if $0 < p \leq 1/3$ **then**
    Predict the noise with $\epsilon_\theta^{scr}(z_t^{scr}, t, c^{scr})$
  else if $1/3 < p \leq 2/3$ **then**
    get $\tilde{z}^{scr}$ using Equation 17
    Predict the noise with $\epsilon_\theta^{scr}(\tilde{z}^{scr}, t, c^{scr})$
  else if $p > 2/3$ **then**
    get mixed sample $\tilde{z}^{scr}$ using RPSC.
    get corresponding language description $\tilde{l}$ and embeded feature $\tilde{c}^{scr}$
    Predict the noise with $\epsilon_\theta^{scr}(\tilde{z}^{scr}, t, \tilde{c}^{scr})$
  end if
  Calculate the gradient descent $\nabla_\theta(\mathcal{L}_{dm} + \mathcal{L}_{con})$
  Update θ with $\theta = \theta - \eta\nabla_\theta$
  Update β using Equation 20
end while

6) Processes of Optimizing and Sampling: To optimize the diffusion model, we take a random hyperspectral patch $\mathcal{H}$ with corresponding text description l, time step t, as well as Gaussian noise ϵ and update the noise prediction model Txt2HSI-LDM(VAE) at each iteration. For the sampling process, we use (14) through the DDIM method to generate the latent space feature z_0, and then, the decoder $\mathcal{D}$ of VAE is used to generate the synthesis image patch.

We summarize the optimization and sampling procedures of the LDM in Algorithms 1 and 2. In Algorithm 1, a variable p is sampled from a uniform distribution every mini-batch to control and select which augmentations (original, LF-UE, and RPSC) need to be used.

E. Classification Stage

1) Expanding the Training Data: In this section, we combine the limited labeled data with synthetic data to train our previously proposed HSIC model SS-ConvNeXt [22]. Unlike using linear interpolation or ROS methods to expand the training data [99], [100], in this work, the sampling process of Txt2HSI-LDM(VAE) is nonlinear and considers the varying degrees of mixing from the perspective of latent feature level and prior language description (i.e., spatial mixup), which confirm that the samples we generated will be more diverse. To better solve the unsatisfactory HSIC performance due to ISSD problem in the real world of remote sensing, one easy but effective way is to generate more data for that small sample, while less data for that larger sample to balance

Algorithm 2 Sampling of Txt2HSI-LDM(VAE)

Input: text description l, the frozen decoder $\mathcal{D}$ of VAE, the frozen Ensemble Latent Diffusion Model ϵ_β^{trg}, guidance coefficient ω

Output: Latent feature $z_0 \sim q(z_0)$ and generated data $\mathcal{H}^{gen}$

Sample Gaussian noise $z_T \in \mathbb{R}^{4\times H\times W}$
for $t = T, \ldots, 1$ **do**
  Calculate ϵ_β
  Sample $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$ if $t > 1$ else $\epsilon_t = 0$
  Calculate z_{t-1} using Equation 14
end for
Generate synthesis hyperspectral patch data $\mathcal{H}^{gen}$ using $\mathcal{D}(z_0)$

the distribution of training data while fully considering the proportion of different categories in the labeled data. Hence, we introduce a scale factor named *sample balance rate (SBR)* λ. The number of generated samples $\tilde{N}$ for specific category is given as follows:

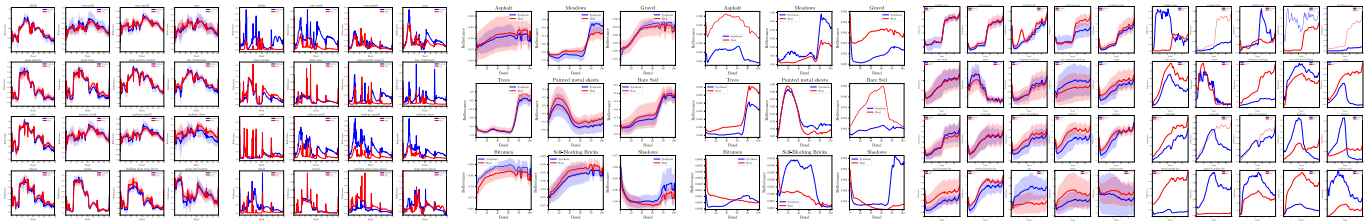
$$r(i) = \lceil \lambda \times \frac{\mathrm{Max}(N)}{N(i)} \rceil, i = 1, \ldots, c$$
$$\tilde{N}(i) = r(i) \times N(i) \tag{22}$$

where N represents the number of each labeled data, c is the number of category, and $\lceil * \rceil$ represents the rounding up to an integer. In our experiment, we set $\lambda = 0.4$.

2) Effectiveness of Generated Samples: To prove the effectiveness when use the generated data to expand the training data, we analyze the characteristics of synthetic samples in two ways.

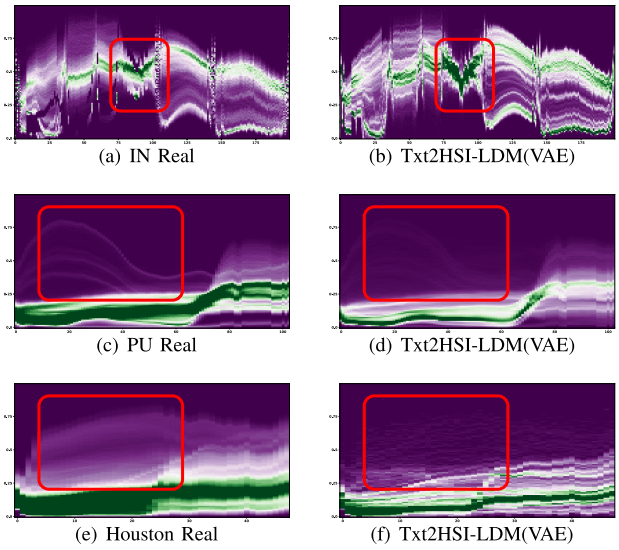
a) Statistical Characteristics of Generated Samples: The error bar is used to depict the changes in mean value and std of generated samples along the spectral dimension. It should be noted that all the results are calculated according to the central pixel of the generated hyperspectral patch. As we can see from Fig. 8, with only limited labeled data when training the language-informed diffusion model, Txt2HSI-LDM(VAE) can fully take advantage of abundant unlabeled data and generate realistic samples, which can be proved from the error bar plot and the change of std along the band dimension. The changes in std on generated samples depict the band-by-band variance characteristics. The error bars of the generated samples are more similar to the real change of all labeled data. Also, in Fig. 9, which shows the spectral curve distributions of the real HSI and the generated HSI for all the labeled data, we stack all the labeled spectral curves of a particular HSI to show the spectral distributions. For the Indian Pines (IN) dataset, we generate the same number of (10 249 samples) spectral curves with this dataset, as well as the Pavia University (PU) dataset (42 776 samples), while for Houston dataset, since the labeled data are more than these two datasets, we fractionally generate a total of 6256 samples. The boundary shows whether the generated spectral curves are consistent with the real HSI, and the density (color) infers the distribution of spectral curves. As shown in Fig. 9, the proposed Txt2HSI-LDM(VAE) generates high-quality spectral curves that have a similar distribution to the real one, also the local pattern, for example, the middle

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(a) (b) (c) (d) (e) (f)

Fig. 8. Statistical characteristics comparison between the generated spectral curves and real ones on Indian Pines dataset, Pavia University dataset, and Houston18 dataset for different land-cover and land use categories. The red solid line represents all the real labeled data, while the blue solid line represents the generated sample. The red and blue shadows represent the std of real and generated samples. (a) Error bar plot on Indian Pines. (b) STD along the spectral dimension on Indian Pines. (c) Error bar plot at Pavia University. (d) STD along the spectral dimension at Pavia University. Zoom in for the best. (e) Error bar plot at Houston18. (f) STD along the spectral dimension at Houston18. Zoom in for the best.



(a) IN Real (b) Txt2HSI-LDM(VAE)
(c) PU Real (d) Txt2HSI-LDM(VAE)
(e) Houston Real (f) Txt2HSI-LDM(VAE)

Fig. 9. Spectral curve distributions of real HSI and generated HSI. All labeled data are plotted. (a) and (b) IN dataset, (c) and (d) PU dataset, and (e) and (f) Houston 2018 dataset. Colored boxes highlight characteristic wavelength regions.

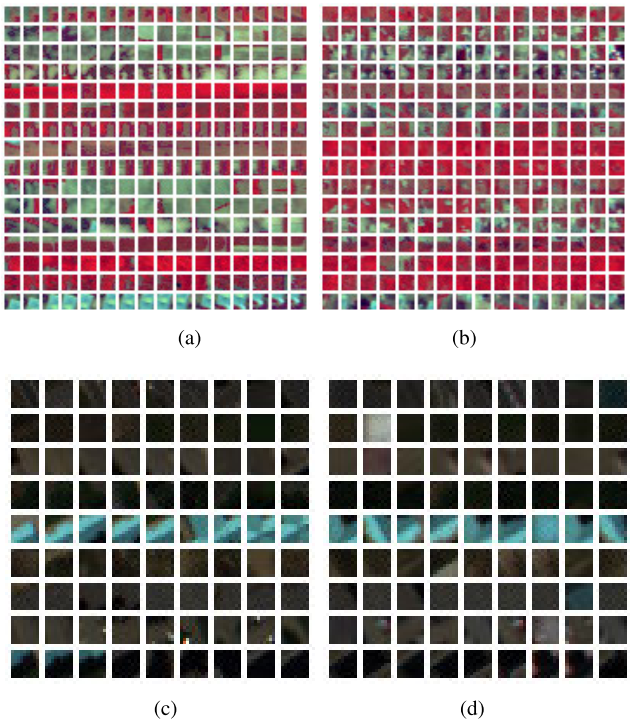
TABLE I

POINT FIDELITY ON EACH GENERATED GROUND COVER TYPE

Class No.	PF (max/min) ↑						
	Dataset			Dataset			
	IN	PU	Houston	Class No.	IN	PU	Houston
1	0.9865/0.8279	0.9984/0.5734	0.9978/0.5128	11	0.9585/0.9585	-	0.8961/0.8961
2	0.9975/0.6190	0.9994/0.7207	0.9996/0.6823	12	0.9543/0.9543	-	0.9794/0.9794
3	0.9974/0.7478	0.9812/0.9812	0.9989/0.9110	13	0.8968/0.8968	-	0.9582/0.9582
4	0.9949/0.8200	0.7361/0.7361	0.9991/0.6924	14	0.7652/0.7652	-	0.9634/0.9634
5	0.8137/0.8137	0.9552/0.9552	0.9994/0.7343	15	0.8499/0.8499	-	0.9587/0.9587
6	0.8831/0.8831	0.8994/0.8994	0.9994/0.7972	16	0.9620/0.9620	-	0.9796/0.9796
7	0.8643/0.8643	0.9781/0.9781	0.9503/0.4887	17	-	-	0.9821/0.9821
8	0.8815/0.8815	0.9796/0.9796	0.9995/0.7588	18	-	-	0.9737/0.9737
9	0.9250/0.9250	0.9364/0.9364	0.9994/0.4242	19	-	-	0.9780/0.9780
10	0.9697/0.9697	-	0.9571/0.9571	20	-	-	0.9670/0.9670

dark green part on the IN dataset. Table I shows the maximum and minimum values of cosine similarity between real patches and generated patches [101]. Txt2HSI-LDM(VAE) achieves the high point fidelity, indicating that its generated images better preserve the spectral content of real images. Some point fidelity of land cover is nearly up to 1. Meanwhile, the minimum value of some land-cover types is around 0.7, meaning that the spectral diversity raised by spatial mixing contributes to more diverse synthetic samples.

b) RGB Visualization of Synthetic Images: Some randomly selected real patch samples and the generated images are shown in Fig. 10. Since Txt2HSI-LDM(VAE) is a transformer-based model, there is no grid overlay on the synthetic image due to the downsampling. Meanwhile, the generated images have good spatial distribution not only with good color visualization, but also with diversity.

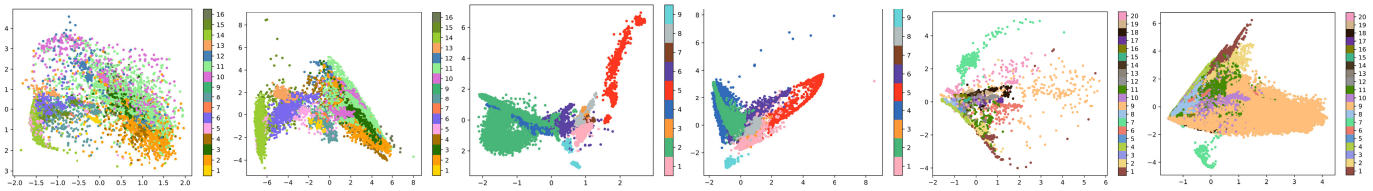


(a) (b)
(c) (d)

Fig. 10. Pseudo-color visualization of (left) real HSI and (right) generated HSI patches with a size of 9×9. Each row represents a specific ground cover. R: 50, G: 30, and B: 10. (a) and (c) Real. (b) and (d) Txt2HSI-LDM(VAE).

c) Data Distribution of Generated Samples in PCA Space: Data reduction is an effective way to analyze the data distribution. One of the popular methods is PCA. We reduce the central pixel of the generated patch data to 2-D space. Predictably, as shown in Fig. 11, when we reduce the spectrum data into 2-D space using PCA, we can observe that the relative location of a specific category in latent space has a high consistency compared to the real data, indicating that the synthetic data generated by Txt2HSI-LDM(VAE) can better capture the data distribution status.

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(a) (b) (c) (d) (e) (f)

Fig. 11. Data distribution of the central pixel of the generated sample in PCA space on Indian Pines and Pavia University datasets. Different colors represent different categories. (a) PCA feature on the generated samples on Indian Pines (10 249 samples). (b) PCA feature on all labeled data on Indian Pines. (c) PCA feature on the generated samples at Pavia University (42 766 samples). (d) PCA feature on all labeled data at Pavia University. (e) PCA feature on the generated samples on Houston (6256 samples). (f) PCA feature on all labeled data at Houston dataset.



(a) (b) (c)

Fig. 12. Pseudo-color image with corresponding ground-truth map and the legend of the datasets. (a) Indian Pines. (b) Pavia University. (c) Houston18.

TABLE II

LAND-COVER TYPES, AND TRAINING AND TESTING SAMPLE NUMBERS OF THREE USED DATASETS. TRAINING SAMPLES REPRESENT THE SAMPLES PER CLASS BEFORE GENERATION

Class No.	Indian Pines			Pavia University			Houston 2018		
	Color	Training Sample	Testing Sample	Color	Training Sample	Testing Sample	Color	Training Sample	Testing Sample
1		3	43		15	6616		3	9555
2		12	1416		43	18606		3	32101
3		8	822		5	2094		3	681
4		2	235		7	3057		3	13585
5		4	479		4	1341		3	5045
6		7	723		11	508		3	4281
7		3	25		4	1326		3	263
8		4	474		8	3674		3	39041
9		3	17		3	944		39	220966
10		8	964					5	45242
11		22	2433					3	33684
12		5	588					3	1513
13		2	203					5	46055
14		11	1254					3	9846
15		3	383					3	6844
16		3	90					3	11396
17								3	146
18								3	6575
19								3	5362
20								3	6821
Total		100	10149		100	42676		100	499102

TABLE III

FINE-GRAINED TEXT DESCRIPTIONS FOR INDIAN PINES DATASET. RED WORDS REPRESENT THE CATEGORY

Class No.	Fine Grained Text	Class No.	Fine-Grained Text
1	Alfalfa occupies a small area and appears light green	9	The shape of oats field is rectangle and has small planting areas
2	Corn notill appears as a large block and has three farming modes	10	The soybean notill fields have not been ploughed
3	The corn mintill is next to corn notill	11	The soybean mintill occupies the largest area and has been lightly ploughed
4	The corn is next to corn mintill and appears irregular in shape	12	The soybean clean is next to stone steel tower and has been harvested
5	The location of grass pasture is relatively dispersed	13	Wheat is next to soybean mintill
6	The area of grass trees is relatively concentrated	14	The woods appears dark green
7	Grass pasture mowed is next to alfalfa and occupies a small part	15	The building grass tress drives is one of the constructions
8	The hay windrowed is used as the food of livestock	16	The stone steel towers are next to soybean clean and appears white in image

IV. EXPERIMENTAL RESULTS

A. Experimental Data

1) Indian Pines Data: This dataset was collected by the AVIRIS sensor over Northwest Indiana, USA. This data consists of 145×145 pixels at a ground sampling distance (GSD) of 20 m and 220 spectral bands covering the wavelength range of 400–2500 nm with a 10-m spectral resolution. In the experiment, 24 water-absorption bands and noise bands were removed, and 200 bands were selected. There are 16 mainly investigated categories in this studied scene. Fig. 12(a) shows the false-color map and ground-truth map. Training and testing numbers are shown in Table II. The fine-grained text description is shown in Table III.

TABLE IV

FINE-GRAINED TEXT DESCRIPTIONS FOR PAVIA UNIVERSITY DATASET. RED WORDS REPRESENT THE CATEGORY

Class No.	Fine Grained Text	Class No.	Fine-Grained Text
1	The asphalt road appears grey and black and there were many trees along the asphalt road	6	There is no grass growing on the surface of bare soil
2	Meadows grows sparsely covered with grass	7	Bitumen is a widely used waterproof material
3	Gravel is a material used in buildings and roads	8	Self-blocking bricks are usually laid flat on the road
4	The trees appear as small circular regions and grow next to the road	9	The shadows is next to the buildings and appear black in color
5	Painted metal sheets are often used as roofing surfaces		

2) Pavia University Data: It was acquired by the ROSIS sensor over Pavia University and its surroundings, Pavia, Italy. This dataset has 103 spectral bands ranging from 430 to 860 nm. Its spatial resolution is 1.3 m, and its image size is 610×340. Nine land-cover categories are covered. Fig. 12(b) shows the false-color map and ground-truth map. Training and testing numbers are shown in Table II. The fine-grained text description is shown in Table IV.

a) Houston 2018 Data: The Houston 2018 dataset was provided by the 2018 IEEE GRSS Data Fusion Contest and acquired by the National Center for Airborne Laser Mapping over the University of Houston campus and its neighborhood. The HSI covers a 380–1050-nm spectral wavelength range with 48 bands at a 1-m GSD of size 601×2384. The ground-truth samples contained 20 classes. Fig. 12(c) shows the false-color map and ground-truth map. Training and testing numbers are shown in Table II. The fine-grained text description is shown in Table V.

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TABLE V
FINE-GRAINED TEXT DESCRIPTIONS FOR HOUSTON18 DATASET. RED WORDS REPRESENT THE CATEGORY

Class No.	Fine Grained Text	Class No.	Fine Grained Text
1	Healthy grass is evenly distributed near the building and appears green, and it is often adjacent to the sidewalk	11	Sidewalks are kind of road and are narrower than roads
2	The stressed grass is not healthy grass and appears light yellow, growing sparsely on the soil.	12	Crosswalks are often in the shape of straight lines and connected with roads and major throughfares
3	Artificial turf is a kind of material that is used for building decoration and stadiums, occupying large areas	13	Major throughfares are much wider than roads and run through the entire area, forming a vertically intersecting shape
4	Evergreen trees usually grow near the buildings and sidewalks, appearing dark green with circular tree canopies	14	Highways are often next to the major throughfares and in the shape of straight lines
5	Deciduous trees have a similar planting distribution as evergreen trees, but leaves will show yellow and green seasonally	15	Railways are usually adjacent to the major throughfares and non-residential areas, appearing white and grey color
6	Bare earth shows brown color and may grows sparse grass	16	Paved parking lots are in the shape of straight lines, and many cars are parked above and near the lots, appearing darker color than unpaved lots
7	Water body has an irregular shape and is located around the vegetation and buildings	17	Unpaved parking lots are under construction and located near the throughfares, showing brighter color
8	Residential buildings are often in the form of regular polygons and are densely clustered together, usually close to sidewalks and roads	18	Cars are often in the shape of rectangle and located near the parking lots, appearing in a densely distributed form
9	Non-residential buildings are usually surrounded by trees and have a similar distribution as residential areas, the roof material usually has a bright color	19	Trains are on the railways with the shape of lines, consisting of many rectangular carriages
10	The roads are usually in the shape of straight lines or curves, connecting the buildings and having trees on both sides	20	Stadium seats are arranged in an overall curved pattern and are closely packed

B. Experimental Setting

1) Implementation and Training Details: Our proposed Txt2HSI-LDM(VAE) is implemented on the PyTorch 1.10.2 platform using a NVIDIA RTX A6000 Ada Generation with 48 GB of VRAM. In the pretraining of VAE, we set epoch to 4000, batch size to 64, and patch size to 9 for all datasets. The epoch and batch size of pretraining of the semi-supervised language-informed diffusion model are 2000 and 64, respectively. The AdamW optimizer is adopted with a learning rate of $1e^{-4}$. In the classification stage, we choose our previous model SS-ConvNeXt [22] with 400 epochs. We calculate the results by averaging the results of ten repeated experiments with 100 training samples selected randomly.

2) Comparing With Other Methods: To compare the effectiveness of the proposed Txt2HSI-LDM(VAE), several representative algorithms are selected for the control experiments, including a conventional classifier: support vector machine (SVM), CNN-based networks: 2D-CNN [3], SSRN [18], SS-ConvNeXt [22], semi-supervised model: DFRes-CR [94], and self-ensembling network: RSEN [95]. Additionally, three diffusion model-based methods, including DKDMN [47], DEMAE [102], and DiffusionAAE [103] are also benchmarked. A linguistic–visual feature alignment method, LDGNet [78], is also included. Since LDGNet is proposed for cross-domain classification, we test by excluding the target-domain dataset, train, and test on the source dataset instead.

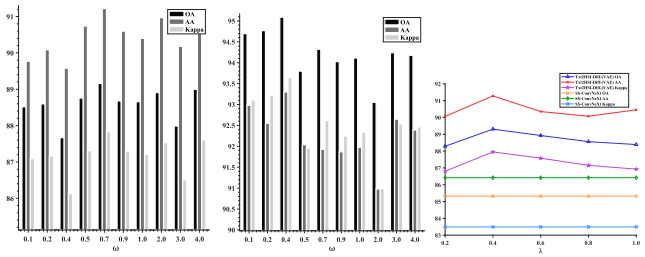
1) *DFRes-CR:* An end-to-end semi-supervised full-convolutional classification model is developed using probability bars for class prediction. The component random field selects similar pixels in the neighborhood of the center pixel to generate pseudo-labels, which are then used to expand the training dataset.
2) *RSEN:* A semi-supervised learning model consists of a base network and an ensemble network. The model jointly constrains the supervised loss of labeled data and the consistency loss of unlabeled data to update model parameters. Consistency filters are also employed to filter high-quality unlabeled data.
3) *DEMAE:* A two-stage model combines masked autoencoder (MAE) and diffusion model to capture the spatial correlation as well as the data distribution under the condition of category label.
4) *DKDMN:* This model establishes two views: one is the original hypersepctral patches, and another is the reconstructed land-cover distribution knowledge, and uses contrastive learning to enhance the network's cross-sample awareness ability.
5) *DiffusionAAE:* This model incorporates spectral similarity constraints and class label guidance into the diffusion process, ensuring the generation of physically realistic synthetic samples, which shows remarkable performance on minority classes. The model is trained in a GAN-style.
6) *LDGNet:* This model uses a dual-stream architecture to extract two levels (coarse-grained and fine-grained) of linguistic features, which are used as a cross-domain shared semantic space. The visual–linguistic alignment is completed by supervised contrastive learning in semantic space. Since this method is proposed for cross-scene HSIC, we set the source and target scenes to the same scene.

To quantitatively evaluate the proposed method and other compared methods, we choose the following commonly used metrics, i.e., overall classification accuracy (OA ↑), average classification accuracy (AA ↑), category accuracy (CA ↑), and kappa coefficient (κ ↑).

C. Experiments on Parameter Setting and Synthetic Sample Effectiveness

1) Parameter Selection on Guidance Coefficient ω: For the generative model, some related researchers find that more diverse and realistic samples used to expand the limited training sample will boost the classification performance [64], [104], [105]. To confirm this and get better ω, we search for the most suitable guide coefficient from 0.1 to 4.0. As shown in Fig. 13(a) and (b), generally, the precision index shows the trend of first rising and then decreasing. For Indian Pines dataset, when $\omega = 0.7$ achieves the best results, while $\omega = 2.0$ ranks second. For Pavia University dataset, the best classification performance appears at $\omega = 0.4$, and then, $\omega = 0.2$.

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(a) (b) (c)

Fig. 13. (a) Parameter experiment of ω on Indian Pines dataset. (b) Parameter experiment of ω on Pavia University dataset. (c) Parameter experiment on SBR λ, and results on Indian Pines dataset.

TABLE VI

INCREASE IN CLASSIFICATION ACCURACY AFTER ADDING TXT2HSI-LDM(VAE) SAMPLES TO DIFFERENT BASELINE MODELS. RESULTS ON INDIAN PINES WITH 100 RAW LABELED DATA

	Methods			
	SVM	2D-CNN	SSRN	SS-ConvNeXt
OA	54.51(+2.43)	83.66(+30.17)	80.63(+15.29)	89.16(+3.84)
AA	57.36(+4.65)	87.92(+32.81)	84.01(+15.72)	91.21(+4.79)
κ	48.65(+3.43)	81.63(+44.72)	78.05(+17.53)	87.83(+4.34)

2) Parameter Selection on SBR λ: Imbalanced data annotation is supposed to be a prevalent problem in the remote sensing community. ISSD HSIC is the main focus of this work. For better expanding and balancing the training samples, the SBR λ is viewed as a simple rule to control the generated number of specific categories based on the sampling frequency. Some related results are shown in Fig. 13(c); compared to the baseline model SS-ConvNeXt, classification performance after expanding training samples gets a big promotion. Even with little λ, in other words, adding fewer generated data will also boost the performance. When $\lambda = 0.4$, three precision indexes show the highest level. The accuracy obtained by expanding samples outperforms the SS-ConvNeXt by 3.98%, 4.86%, and 4.46% in terms of OA, AA, and κ, respectively. Samples after balancing contribute more to the AA, alleviating the ISSD problem in HSIC.

3) Improvement on Other Classification Methods With Generated Samples: To verify that the generated data can also enhance the classification performance of other methods, we perform experiments on SVM, 2D-CNN, SSRN, and SS-ConvNeXt using 100 real training samples with samples generated by Txt2HSI-LDM(VAE) with λ of 0.4 and ω of 0.7. The improvement in classification performance is presented in Table VI on Indian Pines dataset. The red font in Table VI indicates the improvement. By adding more generated samples, alleviating the data scarcity brought by the ISSD problem, making 4.79% improvement in AA on SS-ConvNeXt. The 2D-CNN and SSRN model enhance the richness of each category's spatial–spectral features due to the addition of generated data. This allows the data to better represent the entire data distribution, thereby improving classification performance. The improvement of 2D-CNN by 30.17% in terms of OA demonstrates that the generated data possess both authenticity and diversity. For the traditional method SVM, the classification accuracy was not significantly improved, with only a 2.43% increase in terms of OA. Likely, the generated data do not satisfactorily capture spectral details and variations, making it difficult to improve a classifier that relies solely on spectral information.

TABLE VII

INFLUENCE OF DIFFERENT GRANULARITY TEXT GUIDANCE SIGNALS ON CLASSIFICATION ACCURACY, TAKING INDIAN PINES AS AN EXAMPLE

	No text	Machine generated	Coarse-grained	Fine-grained
OA	85.32±3.10	87.84±0.93	88.38±1.73	89.16±**0.91**
AA	86.42±7.58	90.33±1.66	90.25±1.39	91.21±**1.06**
κ	83.49±3.48	86.18±1.06	86.97±1.94	87.83±**1.02**

4) Impact of Different Granularity of Text Descriptions: In this section, we discuss the level of classification performance contribution on fine-grained and coarse-grained text descriptions. There are two kinds of text designs listed in Section III-D3. The "No text" means directly using SS-ConvNeXt to conduct HSIC without any expanded training sample, while "Coarse-grained" and "Fine-grained" represent using a template and the manually designed text description to generate hyperspectral patches. As we can see from Table VII, not only "Fine-grained" but also "Coarse-grained" will boost the classification performance, and outperform the baseline model by 3.06% and 3.84% in terms of OA, suggesting that linguistic modality enhances the learning of visual representations, and both coarse and fine-grained visual–linguistic similarity calculation via cross-attention can help improve the authenticity of synthetic samples. Additionally, well-designed text improves by around 1% than coarse-grained one, indicating that fine-grained language boosts visual–linguistic alignment.

D. Numerical Evaluation on HSIC

Quantitative assessments of classification performance are presented in Tables VIII–X. The best classification values are colored in shadow, and the optimal std is displayed in bold. Our proposed Txt2HSI-LDM(VAE) expands the training data by generating samples via a language-informed conditional diffusion model to solve ISSD in HSIC. The results show that the proposed Txt2HSI-LDM(VAE) is superior to all other techniques in terms of OA, AA, and κ, and represents better performance on classwise accuracy simultaneously.

1) Performance on Indian Pines: Table VIII shows the quantitative comparison results of Indian Pines dataset. Benefiting from the realistic and diverse generated samples by Txt2HSI-LDM(VAE), the performance of our proposed method is higher than other conventional classifiers, CNN backbone, and semi-supervised networks. For instance, Txt2HSI-LDM(VAE) is 35.67%, 23.82%, 3.84%, 5.3%, and 20.47% better than 2D-CNN, SSRN, SS-ConvNeXt, DFRes-CR, and RSEN in terms of OA. The results show that Txt2HSI-LDM(VAE) can obtain label-domain classification results with higher precision masks that have better raster reversibility under very limited labeled data. In terms of AA, the proposed Txt2HSI-LDM(VAE) exhibits a performance that is superior to DFRes-CR, with an increase of approximately 10.12%. For the small category, the proposed

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TABLE VIII

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, κ, AS WELL AS THE ACCURACIES FOR EACH CLASS ON INDIAN PINES DATASET. THE BEST RESULTS ARE COLORED IN SHADOW, AND THE OPTIMAL STD IS DISPLAYED IN BOLD

Class No.	Color	Conventional Classifier		CNNs Based Networks		Semi-supervised Based Networks		Diffusion Model Based Networks			Language-informed Networks	
		SVM	2D-CNN	SSRN	SS-ConvNeXt	DFRes-CR	RSEN	DKDMN	DEMAE	DiffusionAAE	LDGNet	**Txt2HSI-LDM(VAE)**
1		52.09±17.86	62.95±21.68	84.59±16.16	86.50±16.88	90.91±30.15	85.05±20.13	99.53±0.93	99.30±1.56	92.09±13.56	11.74±14.39	100.00±**0.00**
2		42.32±5.17	44.12±9.33	54.74±12.12	83.87±6.77	79.89±9.42	60.40±9.72	77.45±7.60	71.71±10.07	69.29±7.41	34.67±3.56	79.87±**2.81**
3		40.01±9.12	24.80±9.62	39.73±9.98	76.80±10.65	69.49±16.64	42.63±13.85	74.08±15.47	71.53±9.18	54.01±8.81	38.64±7.26	84.09±**7.46**
4		14.13±5.86	20.46±10.15	39.21±16.72	84.11±7.08	74.45±18.35	11.25±**3.84**	78.63±3.08	64.93±19.81	44.89±13.96	47.24±7.50	85.02±9.43
5		41.21±14.81	23.83±14.71	51.36±16.80	70.38±6.28	50.07±30.99	45.66±15.87	73.77±8.87	77.64±7.89	54.09±13.10	52.13±5.29	74.27±**5.34**
6		84.98±8.21	60.61±20.06	87.31±7.64	95.72±3.34	97.56±3.45	95.30±5.07	95.92±2.49	92.83±5.91	70.38±6.13	76.72±4.75	95.71±1.51
7		89.20±7.55	80.53±19.41	96.50±6.56	100.00±**0.00**	98.76±2.12	99.42±1.51	98.40±3.20	100.00±0.00	97.60±6.31	0.00±0.00	100.00±**0.00**
8		49.14±14.59	77.36±15.22	73.89±14.30	82.21±13.74	98.31±**5.59**	95.54±5.69	88.43±12.07	90.71±7.74	82.15±18.38	93.05±1.48	91.58±13.86
9		91.76±4.96	94.51±13.43	98.52±2.72	99.25±0.23	81.81±4.05	86.55±19.43	100.00±0.00	99.41±1.86	97.64±7.44	0.00±0.00	100.00±**0.00**
10		38.43±9.88	42.72±11.58	51.93±10.94	79.50±**0.57**	79.64±7.28	48.53±19.43	85.22±5.94	67.76±4.41	68.79±6.87	59.03±4.47	88.76±7.09
11		61.61±3.51	71.30±6.63	78.87±4.55	89.71±4.01	94.60±3.50	86.87±6.19	85.28±4.05	82.52±5.75	74.40±2.52	59.03±4.47	93.43±**3.35**
12		22.52±7.88	18.21±8.58	25.57±**5.53**	75.56±15.15	68.23±19.96	20.07±8.51	73.07±10.52	57.19±15.77	51.29±11.41	40.19±5.51	86.62±9.36
13		77.04±17.75	65.94±25.91	81.12±19.18	95.23±5.74	99.86±0.23	81.02±25.51	96.15±4.54	98.07±3.40	86.99±10.46	92.18±7.33	99.90±**0.22**
14		71.88±4.46	79.97±8.24	91.31±6.66	95.42±7.03	99.51±**1.32**	92.47±10.50	98.69±6.28	98.11±1.35	86.99±4.62	91.98±0.35	96.46±1.98
15		17.00±6.92	26.23±16.27	43.92±8.13	64.24±11.11	36.43±15.89	46.44±25.38	79.26±6.56	56.21±15.18	58.45±12.12	39.88±2.82	83.74±**2.83**
16		50.11±21.69	82.22±6.77	94.02±7.10	97.13±5.64	93.83±3.01	72.06±34.26	97.77±2.81	97.56±6.57	96.66±6.43	76.85±4.44	100.00±**0.00**
OA(%)		52.08±1.91	53.49±2.11	65.34±3.91	85.32±3.10	83.86±2.39	68.69±1.75	84.58±2.90	79.25±2.44	70.51±3.24	63.34±1.31	89.16±**0.91**
AA(%)		52.71±2.12	55.11±29.88	68.29±10.30	86.42±7.58	82.09±13.02	67.23±2.78	87.60±2.60	82.85±1.48	74.20±3.52	51.98±30.24	91.21±**1.06**
κ(%)		45.22±2.38	46.91±2.69	60.52±4.59	83.49±3.48	81.57±2.81	65.53±1.97	82.50±3.32	76.34±2.76	66.59±3.89	57.82±1.61	87.83±**1.02**

TABLE IX

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, κ, AS WELL AS THE ACCURACIES FOR EACH CLASS ON PAVIA UNIVERSITY DATASET. THE BEST RESULTS ARE COLORED IN SHADOW, AND THE OPTIMAL STD IS DISPLAYED IN BOLD

Class No.	Color	Conventional Classifier		CNNs Based Networks		Semi-supervised Based Networks		Diffusion Model Based Networks			Language-informed Networks	
		SVM	2D-CNN	SSRN	SS-ConvNeXt	DFRes-CR	RSEN	DKDMN	DEMAE	DiffusionAAE	LDGNet	**Txt2HSI-LDM(VAE)**
1		74.87±5.49	85.24±9.82	89.69±4.23	94.68±3.37	84.60±8.93	95.42±**2.72**	94.05±2.63	70.65±13.08	78.79±3.72	80.81±3.54	93.34±5.77
2		88.81±3.65	95.45±2.50	96.12±2.24	97.00±2.00	95.67±3.73	98.74±1.34	99.85±0.19	94.65±9.43	97.39±1.30	96.32±1.33	99.08±**0.77**
3		45.15±12.12	24.92±16.98	54.83±17.92	70.43±14.32	52.12±20.72	48.06±24.63	68.53±5.94	89.81±8.36	70.97±4.50	68.61±11.68	81.98±**11.09**
4		55.57±9.91	82.58±11.98	81.44±5.38	92.48±**4.32**	83.54±4.84	94.06±10.95	77.87±5.13	74.51±7.04	74.99±11.40	89.85±5.72	91.07±4.52
5		48.59±20.42	99.66±0.34	96.85±5.30	99.85±0.31	97.03±2.15	82.57±11.42	100.00±0.00	99.72±0.60	93.25±5.81	98.81±0.78	99.80±**0.16**
6		48.42±8.54	31.76±11.02	81.90±10.03	82.74±7.60	40.50±3.40	61.58±7.32	94.63±3.13	96.35±3.22	99.05±1.03	63.10±10.10	94.61±**2.69**
7		56.37±18.04	55.45±21.79	86.00±12.26	91.09±6.76	58.71±41.48	24.10±10.72	99.03±1.18	93.79±5.96	96.01±0.80	59.23±9.95	91.43±**4.88**
8		68.36±10.67	77.49±11.06	87.27±7.60	93.43±**6.25**	76.38±21.33	85.18±16.16	92.93±2.96	66.66±12.26	80.22±4.36	64.64±8.67	90.48±7.76
9		33.31±20.07	93.53±4.17	97.86±1.37	94.28±3.12	99.79±**0.17**	61.20±38.41	88.29±4.07	84.35±9.61	52.86±12.78	90.09±4.38	97.84±1.23
OA(%)		72.18±2.68	79.29±1.80	89.36±1.99	92.88±1.46	81.78±2.38	86.20±1.97	94.32±0.74	86.94±5.07	89.16±0.49	84.40±1.59	95.09±**1.02**
AA(%)		57.71±3.48	71.79±9.96	85.78±3.22	90.66±**1.44**	76.48±2.41	72.36±4.61	90.57±1.40	85.61±2.67	82.61±2.28	79.06±2.33	93.30±4.32
κ(%)		62.40±3.50	71.71±2.45	85.87±2.65	90.54±1.95	75.30±3.08	81.31±2.65	92.42±1.01	82.99±6.13	85.68±0.67	78.89±2.08	93.65±**1.32**

TABLE X

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, κ, AS WELL AS THE ACCURACIES FOR EACH CLASS ON HOUSTON DATASET. THE BEST RESULTS ARE COLORED IN SHADOW, AND THE OPTIMAL STD IS DISPLAYED IN BOLD

Class No.	Color	Conventional Classifier		CNNs Based Networks		Semi-supervised Based Networks		Diffusion Model Based Networks			Languange-informed Networks	
		SVM	2D-CNN	SSRN	SS-ConvNeXt	DFRes-CR	RSEN	DKDMN	DEMAE	DiffusionAAE	LDGNet	**Txt2HSI-LDM(VAE)**
1		84.39±14.81	79.46±13.27	79.45±22.99	65.22±7.14	89.03±**0.83**	60.83±9.55	74.13±14.05	75.43±13.52	67.35±18.20	59.46±16.08	93.83±4.64
2		58.60±20.56	58.82±16.97	65.51±9.92	82.40±10.75	88.10±**1.25**	39.39±9.16	48.27±4.66	46.62±14.00	28.33±9.00	90.56±2.75	62.56±11.52
3		98.74±1.29	96.10±4.42	92.38±4.35	94.89±2.65	100.00±**0.00**	62.15±16.96	96.87±1.86	99.98±0.04	73.30±13.89	24.93±43.19	99.19±0.45
4		65.90±9.49	79.13±7.43	87.01±7.99	77.57±16.71	97.69±**0.36**	85.26±6.01	81.17±1.94	86.21±8.58	72.77±11.78	75.62±7.83	84.61±6.71
5		51.08±9.88	45.91±13.18	39.61±9.86	56.87±10.01	84.28±**0.38**	18.37±5.88	69.13±4.81	66.21±10.03	38.45±6.65	11.42±9.55	60.96±5.22
6		56.94±**3.59**	52.54±14.01	70.84±9.38	71.75±18.97	87.16±13.99	84.48±8.55	65.90±18.16	69.57±20.00	86.79±7.71	7.27±5.72	73.81±5.76
7		70.19±20.47	96.69±1.92	55.15±22.66	98.92±1.01	100.00±**0.00**	45.96±9.82	100.00±**0.00**	86.00±17.38	53.76±11.37	0.00±0.00	100.00±**0.00**
8		55.01±6.26	19.73±10.77	35.90±17.50	37.40±14.16	80.70±3.50	42.65±5.92	69.00±10.09	47.35±15.52	44.16±6.39	74.76±**2.35**	66.71±11.22
9		46.66±8.22	83.57±4.17	84.88±3.86	37.40±14.16	42.97±7.42	94.54±4.12	89.16±3.01	93.32±2.60	83.49±3.95	79.403±5.05	83.46±**2.08**
10		26.31±6.49	19.75±10.73	23.36±6.01	88.84±2.05	28.02±7.50	1.4±**0.67**	31.13±9.53	26.18±7.62	19.99±9.32	36.88±6.54	15.53±3.73
11		12.80±5.10	8.19±3.12	7.64±2.07	27.05±14.55	34.68±13.26	2.91±**0.60**	16.87±1.10	66.66±12.26	10.59±4.05	26.79±4.68	43.70±3.79
12		26.31±6.49	12.94±8.25	13.68±6.16	13.72±5.43	40.21±6.59	3.54±**1.05**	31.82±10.4	26.89±10.13	14.11±3.14	0.00±0.00	41.87±4.07
13		19.97±8.28	33.64±9.72	33.24±5.35	5.50±4.17	48.72±5.36	7.61±**1.21**	37.43±8.07	23.10±7.86	27.47±4.57	53.46±8.31	42.32±5.95
14		62.78±13.46	40.91±11.59	63.25±9.25	53.82±15.60	94.44±**1.34**	68.51±9.49	70.51±9.91	76.48±10.51	69.08±14.40	6.37±3.92	76.50±9.58
15		69.48±11.12	40.19±9.69	58.00±15.12	61.81±13.31	99.59±**0.20**	96.47±0.68	69.98±15.07	76.76±15.81	76.43±18.61	47.16±28.11	90.54±5.75
16		51.66±18.09	39.08±18.33	38.78±9.32	66.01±18.16	84.81±8.57	43.31±**4.01**	60.09±9.96	60.63±17.12	42.56±7.96	41.04±22.84	72.19±15.49
17		96.96±2.51	98.60±1.92	77.76±18.65	67.02±11.48	100.00±**0.00**	37.60±5.29	100.00±**0.00**	97.73±6.55	92.05±14.20	0.00±0.00	100.00±**0.00**
18		23.78±12.60	36.86±14.03	15.42±4.01	98.18±**2.98**	88.04±3.24	34.59±4.09	49.52±20.58	66.01±20.46	54.12±7.93	11.48±9.28	72.55±6.16
19		35.49±14.74	45.57±17.17	41.01±10.74	16.85±10.84	88.38±7.33	48.44±5.45	88.31±**0.24**	72.22±14.85	68.86±19.82	9.88±6.03	76.96±8.33
20		67.76±13.49	50.16±19.64	55.57±11.78	45.61±18.58	98.73±**0.84**	61.20±38.41	85.85±6.48	85.42±12.32	73.10±7.52	1.55±0.16	89.53±7.16
OA(%)		43.54±3.69	56.55±2.91	59.82±0.71	66.17±1.65	55.21±5.01	58.16±2.33	67.11±1.19	65.80±1.59	57.77±2.85	62.27±1.48	67.74±**1.12**
AA(%)		54.03±1.96	51.89±10.52	52.02±10.35	60.51±10.52	78.77±4.23	72.36±4.61	66.76±1.07	65.00±3.05	54.84±3.61	32.90±21.18	71.39±**1.50**
κ(%)		35.12±3.03	43.66±3.83	47.44±0.60	55.79±2.70	49.13±4.83	41.08±4.23	57.78±0.99	55.10±2.66	46.96±3.26	32.90±4.05	58.26±**1.57**

Txt2HSI-LDM(VAE) outperforms the semi-supervised RSEN by 14.95%, 0.58%, 13.45%, and 27.94% on class 1, 7, 9, and 16, respectively. DFRes-CR only considers the CRF model in the local domain, and the pseudo-label generated by DFRes-CR relies heavily on baseline performance. Although RSEN performs data reduction before performing the classification task, contributing to less training time, but the information loss is also an issue worth considering. After expanding the sample according to the SBR λ, there is little fluctuation in accuracy between the 16 categories. The maximum fluctuation amplitude has been reduced from 35.76% (classes 7 and 15) in the baseline model SS-ConvNeXt to 25.73% (classes 6 and 5),

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TABLE XI
COMPUTATIONAL COST AND MODEL SIZE ACROSS DIFFERENT ARCHITECTURES

Dataset	Metrics	Methods					
		Spectraldiff	DKDMN	DEMAE	DiffusionAAE	LDGNet	Txt2HSI-LDM(VAE)
Indian Pines	FLOPS	1.55T	88.157G	68.595M	36.02G	4.538G	6.39G
	Params	4.300M	4.300M	783.288K	68.77M	36.68M	115.32M
	Training Time	-	2s/epoch	1s/epoch	1h/epoch	1s/epoch	8s/epoch
	Generation Speed	-	-	-	35 IMGs/s	-	16 IMGs/s
Pavia University	FLOPS	809.604G	45.841G	68.595M	17.35G	4.456G	6.39G
	Params	4.300M	4.300M	783.288K	68.77M	33.87M	72.19M
	Training Time	-	9s/epoch	1s/epoch	1h/epoch	1s/epoch	6s/epoch
	Generation Speed	-	-	-	35 IMGs/s	-	16 IMGs/s
Houston	FLOPS	4.110T	21.143G	68.595M	10.78G	4.41G	6.39G
	Params	4.300M	4.300M	783.288K	68.77M	33.65M	115.33M
	Training Time	-	17s/epoch	1s/epoch	2.4h/epoch	3s/epoch	6s/epoch
	Generation Speed	-	-	-	35 IMGs/s	-	16 IMGs/s

Note: Generation speed means how many images are generated in one second.

while the fluctuation amplitude of SVM, 2D-CNN, SSRN, DFRes-CR, and RSEN is 77.63%, 74.05%, 70.93%, 63.43%, and 88.17%, respectively. Hence, the language-informed diffusion model can better capture the linguistic–visual similarity on the pixel level, generating more diverse samples to alleviate the ISSD problem in HSIC. DKDMN uses the contrastive learning method to enhance the discriminative capacity of the model between diffusion knowledge and real samples, making this network achieve 87.60% in terms of AA. Although DEMAE uses a masked autoencoder to learn the spatial coherence, the dimension reduction PCA leads to information loss. DiffusionAAE is trained in GAN-style, which is time-consuming during model pretraining (see Table XI), does not provide more model generalization ability, and only achieves 70.51% in terms of OA, 18.65% lower than our model. LDGNet is proposed to solve the cross-scene HSIC. In order to adapt to a single scene, we test LDGNet on the same image. Although this model used the language of two particle sizes of thickness, the unlabeled data have not been fully used, leading to lower generalization, 39.23 lower than ours in terms of AA.

2) Performance on Pavia University: Table IX illustrates the comparison of the experimental results to various classes on Pavia University dataset. From Table IX, Txt2HSI-LDM(VAE) achieves the best accuracy (OA of 95.09%, AA of 93.30%, and κ of 93.65%) among the methods. The AA obtained by Txt2HSI-LDM(VAE) on Pavia University dataset outperforms SSRN, SS-ConvNeXt, DFRes-CR, and RSEN by 7.52%, 2.64%, 16.82%, and 20.49%, respectively. The maximum fluctuation amplitude has been reduced from 29.426% (classes 2 and 5) in SS-ConvNeXt to 17.82% (classes 3 and 5), while the fluctuation amplitude of SVM, 2D-CNN, SSRN, DFRes-CR, and RSEN is 55.5%, 74.74%, 43.03%, 59.29%, and 74.64%, respectively.

3) Performance on Houston: Table X illustrates the comparison of the experimental results to various classes on Houston dataset. From Table X, Txt2HSI-LDM(VAE) achieves the best accuracy (OA of 67.74% and κ of 58.26%) among the methods. The OA obtained by Txt2HSI-LDM(VAE) on Houston dataset outperforms DKDMN, DEMAE, DiffusionAAE, and LDGNet by 0.63%, 1.94%, 9.97%, and 5.47%, respectively.

E. Visual Evaluation on HSIC

The classification maps are illustrated in Figs. 14–16. In Fig. 14, the proposed Txt2HSI-LDM(VAE) obtains less noisy and more accurate results in some areas of the classification maps, such as the category in the blue block (Corn-notill) and soybean-notill in the yellow block. It is obvious that the boundary of land cover in the red block is better predicted, while other methods blur the boundaries, such as DKDMN and DEMAE. Some related results can also be seen in Fig. 15, such as the details in the yellow block. Due to the combination of CRFs and region growing, DFRes-CR can obtain high-confidence samples and get better classification maps in Fig. 14(e). On more larger scale dataset, Houston, our proposed models can well capture the building or artificial structure boundary (the blue and magenta boxes). Overall, the classification map keeps consistency with the ground truth, representing the scalability of Txt2HSI-LDM(VAE) on a large dataset is promising, which can be explained by the use of unlabeled data as well as the spatial mixing operation during model pretraining, i.e., LF-UE and RPSC.

F. Visualization of Visual–Linguistic Cross-Attention

We return to our key observation: the spatial layout and geometry of the generated image depend on the cross-attention maps. The interaction between pixel and text is illustrated in Fig. 17, where the average attention maps are plotted. Note that averaging is done for visualization purposes, and attention maps are kept separate for each head. As we can see, pixels show a stronger correlation with the words that describe them, e.g., pixels of the *Gravel* are correlated with the word "Gravel," as well as "Painted metal sheets." Additionally, some words related to the category are also captured, such as "roof surface" and "road," while attention maps of prepositions and conjunctions, such as "with" and "on," seem to fail to be

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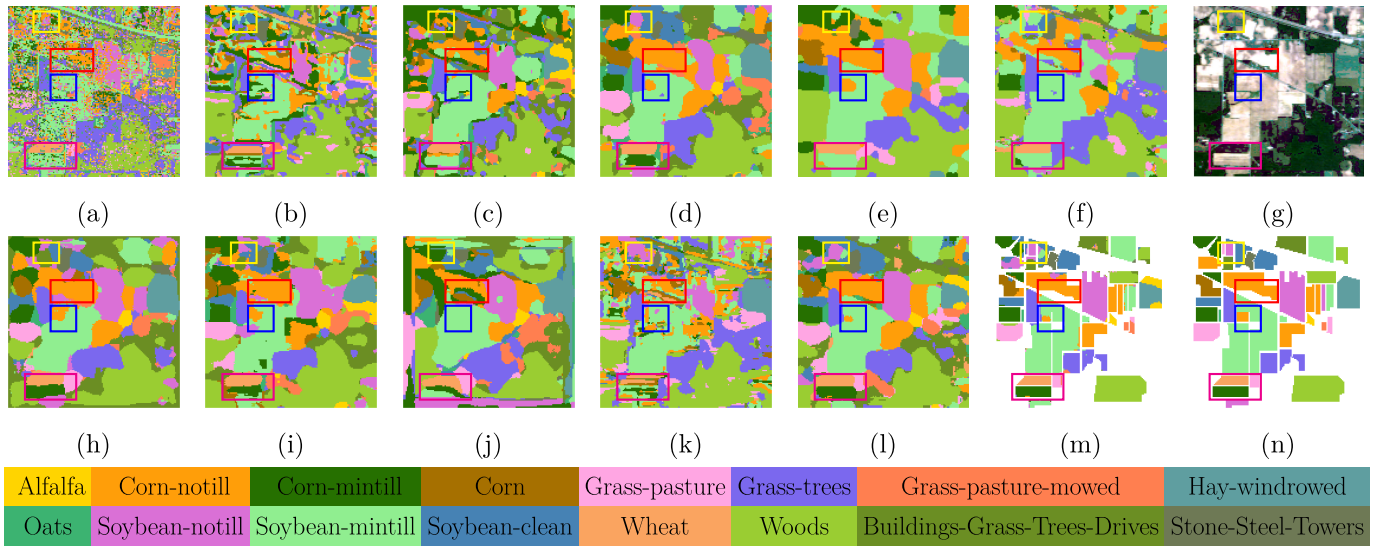


Fig. 14. Classification maps of Indian Pines dataset. (a) SVM (OA = 52.08%). (b) 2D-CNN (OA = 53.49%). (c) SSRN (OA = 65.34%). (d) SS-ConvNeXt (OA = 85.32%). (e) DFRes-CR (OA = 83.86%). (f) RSEN (OA = 68.69%). (g) RGB composite image. (h) DKDMN (OA = 84.58%). (i) DEMAE (OA = 79.25%). (j) DiffusionAAE (OA = 70.51%). (k) LDGNet (OA = 61.34%). (l) Txt2HSI-LDM(VAE) (OA = 89.16%). (m) Label-domain classification map of Txt2HSI-LDM(VAE). (n) Ground-truth map. The box in red, blue, yellow, and magenta shows the region of interest, which represents the category boundary-preserving capability of different methods.

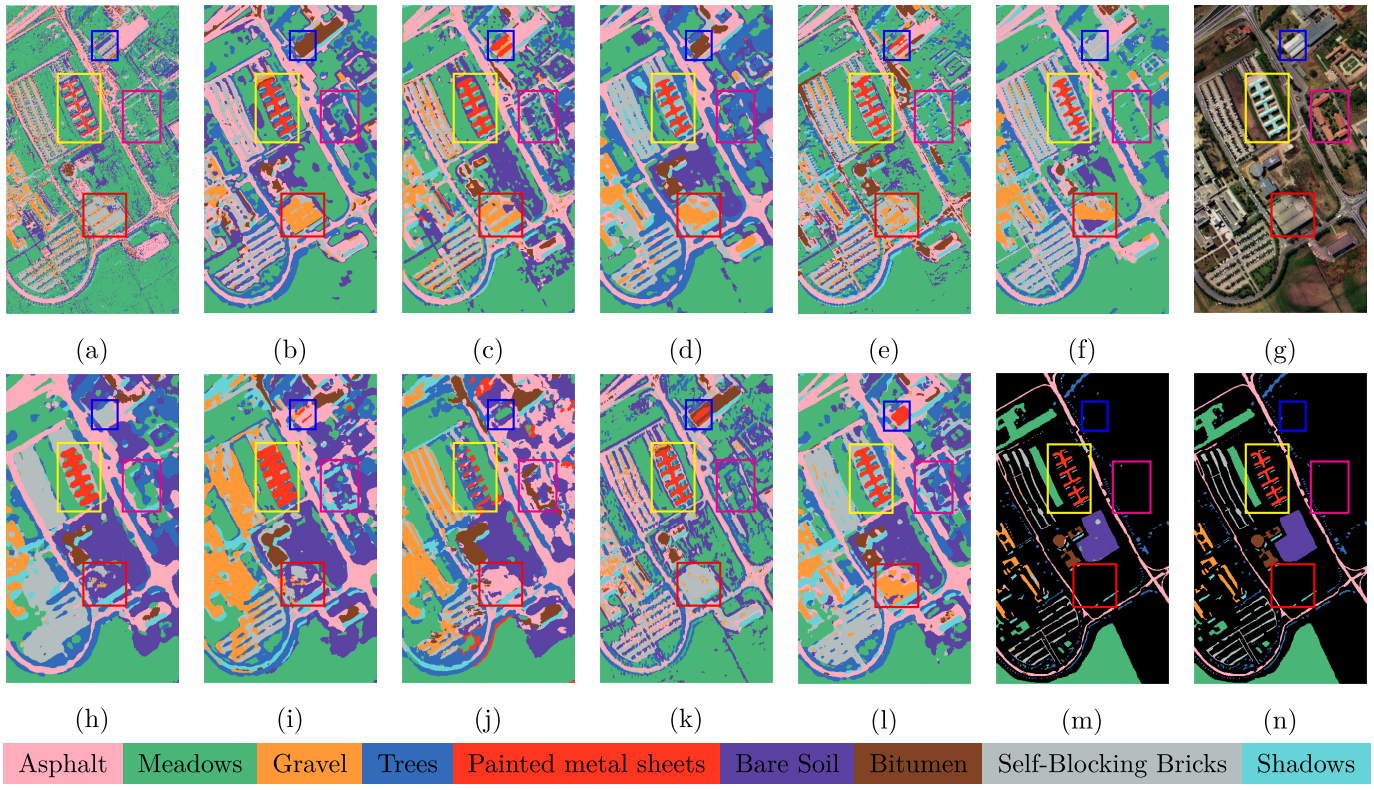


Fig. 15. Classification maps of Pavia University dataset. (a) SVM (OA = 72.18%). (b) 2D-CNN (OA = 79.29%). (c) SSRN (OA = 89.36%). (d) SS-ConvNeXt (OA = 92.88%). (e) DFRes-CR (OA = 81.78%). (f) RSEN (OA = 86.20%). (g) RGB composite image. (h) DKDMN (OA = 94.32%). (i) DEMAE (OA = 86.94%). (j) DiffusionAAE (OA = 89.16%). (k) LDGNet (OA = 84.40%). (l) Txt2HSI-LDM(VAE) (OA = 95.09%). (m) Label-domain classification map of Txt2HSI-LDM(VAE). (n) Ground-truth map. The box in red, blue, yellow, and magenta shows the region of interest, which represents the category boundary-preserving capability of different methods.

captured. These words contain much information about spatial layout and topological relations. In the future, the ability to model prepositions and conjunctions needs to be further strengthened.

G. Computational Complexity

To compare with the computational cost and model size of the diffusion training part, we set the batch size equal to 2 for Spectraldiff, DKDMN, DEMAE, DiffusionAAE, and

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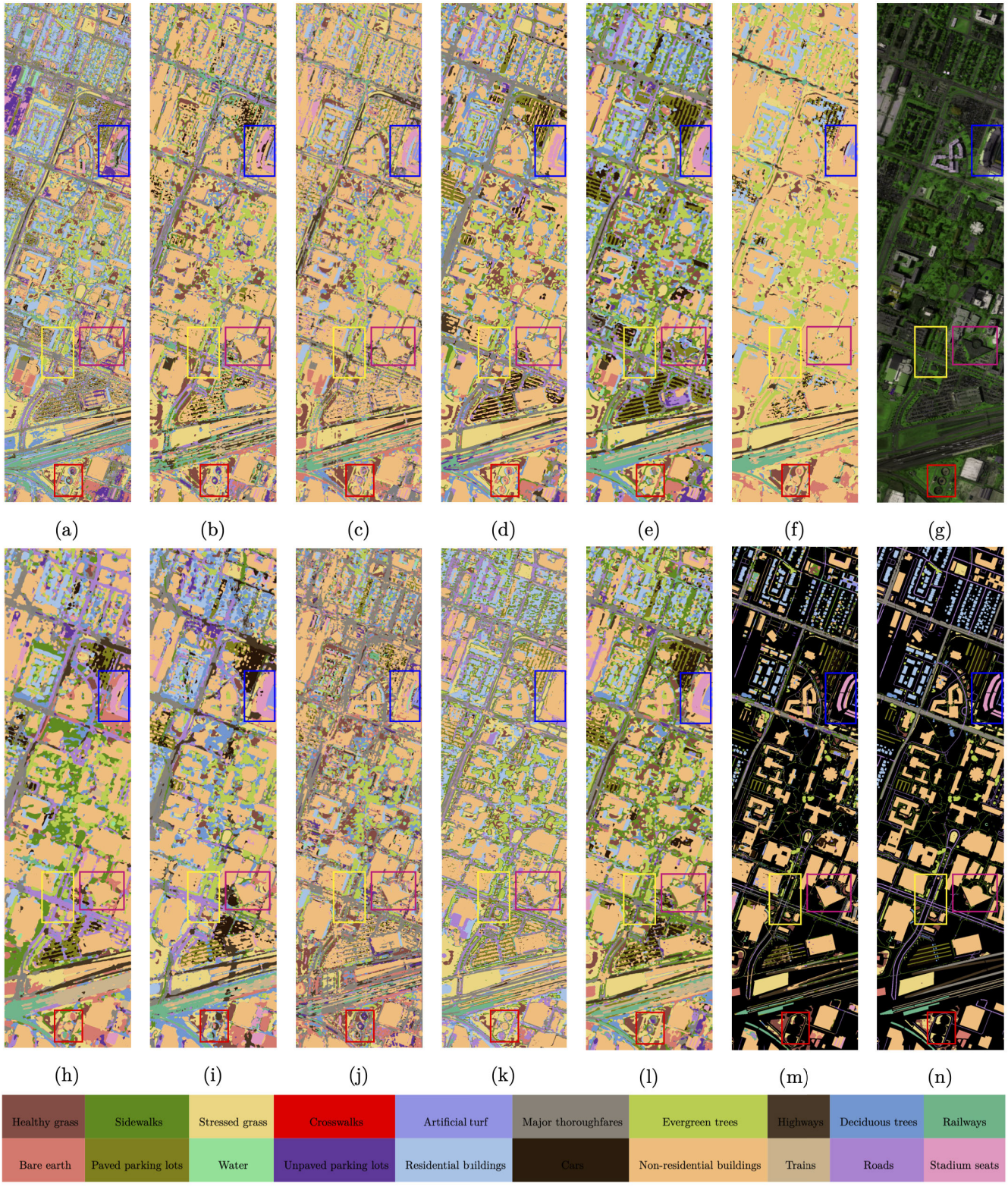


Fig. 16. Classification maps of Houston dataset. (a) SVM (OA = 43.54%). (b) 2D-CNN (OA = 56.55%). (c) SSRN (OA = 59.82%). (d) SS-ConvNeXt (OA = 66.17%). (e) DFRes-CR (OA = 55.21%). (f) RSEN (OA = 58.16%). (g) RGB composite image. (h) DKDMN (OA = 67.11%). (i) DEMAE (OA = 65.80%). (j) DiffusionAAE (OA = 57.77%). (k) LDGNet (OA = 62.27%). (l) Txt2HSI-LDM(VAE) (OA = 67.74%). (m) Label-domain classification map of Txt2HSI-LDM(VAE). (n) Ground-truth map. The box in red, blue, yellow, and magenta shows the region of interest, which represents the category boundary-preserving capability of different methods.

our proposed method, Txt2HSI-LDM(VAE), to calculate the number of parameters as well as the floating point operations (FLOPs). As we can see from Table XI, Spectraldiff has a huge amount of FLOPs, for example, with the shape of [2, 1, 200,

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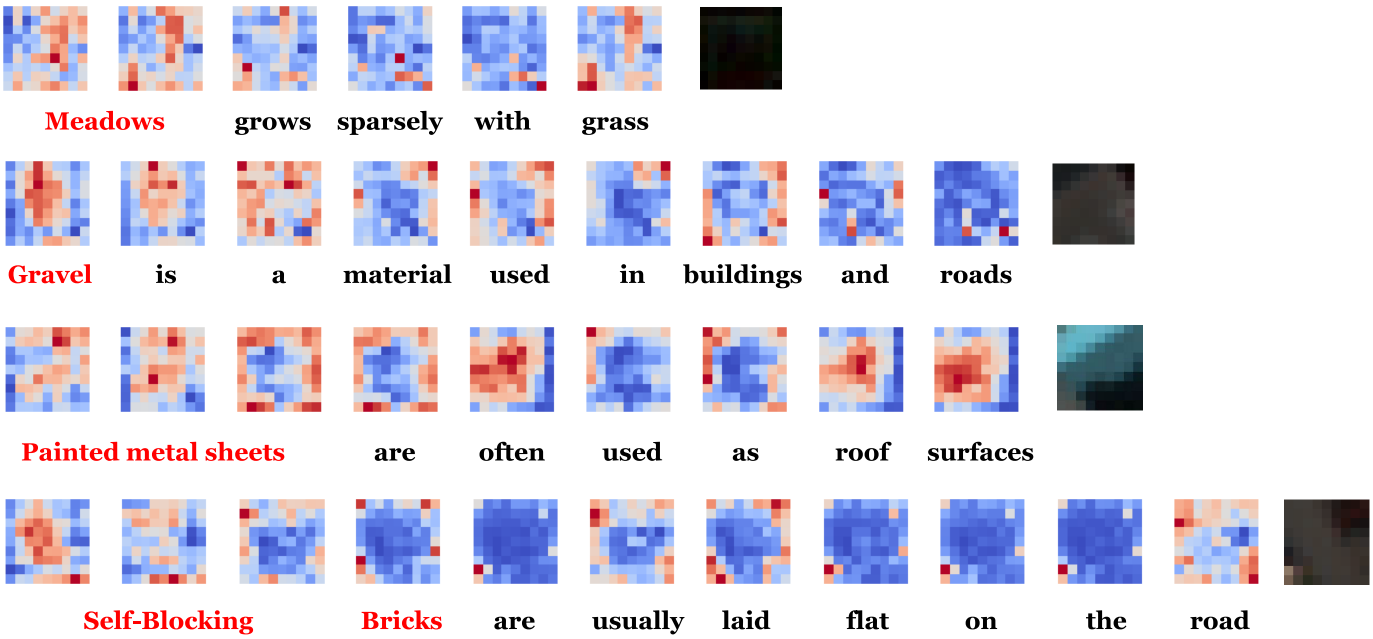


Fig. 17. Average attention masks for each word in the text, which was used to synthesize the right image, showing the similarity of text–image features based on the cross-attention mechanism. The redder the color, the higher the similarity between the language and the spatial pixel. Take Pavia University data as an example. The category names are indicated in red font.

64, 64] as model input, the FLOPs will reach 1.557T, leading to slow model training and inference. During model training, Spectraldiff will cost around seven days to pretrain, which is much longer than Txt2HSI-LDM(VAE), so we exclude this method in comparative experiments and do not include the training time and generation speed here. DKDMN is also implemented based on 3-D convolution, so the computational cost is also higher than Txt2HSI-LDM(VAE). Additionally, due to the PCA reduction, DEMAE reduces all four datasets into 36 dimensions, which hugely reduces the model parameter as well as FLOPs. Our model also has relatively lower FLOPs and model parameters. DiffusionAAE is trained in GAV-style, which needs more pretraining time than other models. The 1 s/16 IMGs means that generating 16 images will cost 1 s.

V. Conclusion and Discussion

The Txt2HSI-LDM(VAE), which uses the condition of language to generate diverse samples for expanding training data, is proposed. The visual–linguistic alignment is established by the cross-attention mechanism. For better learning the spatial–spectral information, we design a semi-supervised conditional diffusion model framework to realize the information interaction between limited labeled data and a bunch of unlabeled data via knowledge distillation. Experiments show that whether it is a fine-grained text or a coarse-grained text, it will promote the classification performance. The spatial-level attention maps show that cross-attention can capture the spatial layout and location of a specific category.

In this work, the text description is designed by hand according to prior knowledge or based on templates. It may be possible to build a specialized text library by involving fine-tuning techniques on ChatGPT, DeepSeek, or developing domain-specific models to realize it, even a large model in the remote sensing field. Additionally, prepositions and conjunctions in text description are especially important to establish reasonable topological relationships of land cover, which need further research. Some research has explored the geospatial reasoning ability of LLMs, such as obtaining coordinates and locations for cities and a quantitative understanding of spatial prepositions by utilizing using the proper "querying" techniques [106]. In text-to-image generation task, exploring and pairing the relations between text and object location in the image may need spatial-relation-aware text encoders and a coordinate-based new architecture. Also, in the cross-attention module part, how to emphasize the importance of prepositions and conjunctions, which will affect the geometric relationship modeling. The most relevant descriptions and significant content should not be submerged in redundant text when building the linguistic descriptions [107]. Moreover, the VAE training objective function is often on the pixel level, lacking constraints on the objective level and explicit geometric consistency. How to avoid or reduce the suppression of high-frequency information, like the boundary and linear feature, also deserves to be studied.

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